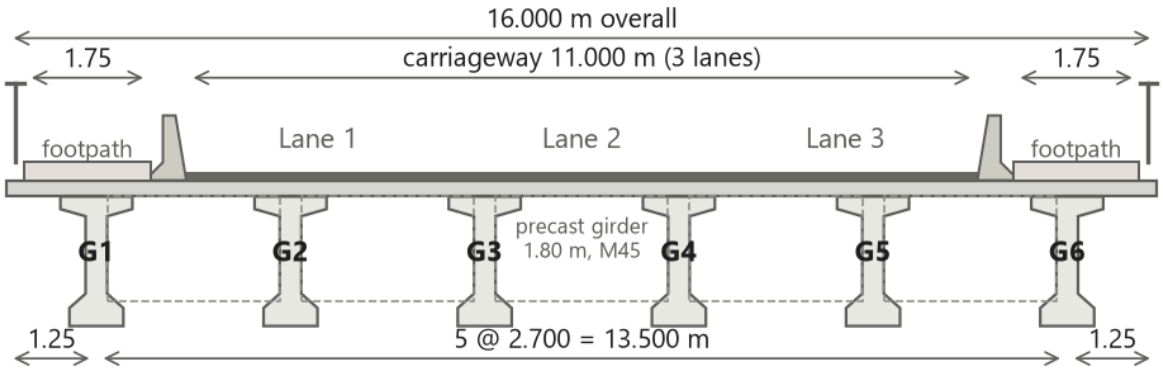
A 33 m post-tensioned PSC I-girder bridge, from design to a maintenance decision

Analysis and design, prestressing, healthy baseline, inspection, NDT and monitoring, synthetic data, AI/ML-assisted assessment and maintenance of Bridge B33



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Summary

This report follows one bridge through its life. Bridge B33 is a three-span bridge with simply supported spans of **33 m**, a 16 m deck carrying three traffic lanes with footpaths and crash barriers on both sides, and **six post-tensioned M45 I-girders at 2.70 m** under a 225 mm deck slab. Every step from configuration to the maintenance plan was worked out for this one span and is carried forward to the next step, so that the measurements, the synthetic data and the AI/ML results all refer to the same structure.

Item	Result
Configuration	6 girders, 1.80 m deep (L/D = 16.3 with slab), end diaphragms + one cross girder at mid-span; elastomeric bearings 400 x 500 mm; strip-seal joints
Live load (IRC:6)	70R wheeled + Class A governs; outer girder LL moment 2426 kN m by grillage, 2457 kN m by Courbon
Prestress	4 cables 12T15 per girder, two-stage stressing; P_j = 9.37 MN, P_0 = 8.37 MN, P_e = 7.17 MN (total loss 23.5 %)
Checks (IRC:112)	rare-combination soffit stress -2.27 MPa (> $-f_{ctm}$ = -3.3, uncracked); quasi-permanent 2.68 MPa (decompression satisfied); M_{Ed}/M_{Rd} = 0.76; LL deflection 11.8 mm (L/800 = 41 mm)
Healthy baseline	f_1 = 3.74 Hz (bending), f_2 = 3.93 Hz (torsion) at 25 °C, -0.15 %/°C; load-test deflection 6.31 mm under 4 x 300 kN lorries
Monitoring	12 accelerometers (layout fixed by an AutoMAC check), 4 VW gauges, 4 bearing displacement transducers, levelling points, temperature; 30 min records at 100 Hz
Synthetic data	3600 campaigns from the updated digital twin: healthy and six deterioration mechanisms, one or two at a time, with temperature, traffic and instrument noise
AI/ML	novelty detection, random forests per mechanism, Gaussian-process estimate of prestress loss, girder localisation, NDT clustering; macro-F1 0.32 with vibration alone and 0.90 with all sources fused
Year-18 decision	Serious - detailed investigation: leaking joint at pier P2 has led to restrained bearings and corrosion at the G1 anchorage end; joint, bearings and concrete repair, tendon investigation and a prestress check, then re-baseline

Three findings carry through the study and are, in my view, the useful part of it. First, the natural frequencies of this bridge change by about 4 % between a cool winter night and a hot afternoon, which is more than most of the deterioration cases change them, so frequencies are only useful after temperature and traffic normalisation. Second, loss of prestress does not change the stiffness of an uncracked bonded girder; it shows in levelling and in vibrating-wire strain long before it shows in vibration or a load test, and the cracking that finally makes it visible starts at roughly 25 % extra loss. Third, the machine-learning models were only as good as the physics in the features given to them: fusing vibration with static, geometric, bearing and NDT data raised the detection quality much more than any choice of algorithm did.

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1 Objective, approach and assumed site

The brief (§1.1) asks for a post-tensioned PSC I-girder bridge to be studied as an infrastructure system over its service life: analysed and designed, prestressed, given a healthy baseline, inspected, tested and monitored, deteriorated in a controlled way, and finally assessed with AI/ML so that a maintenance decision can be made. The chain below is the one I followed. Each block hands a definite set of numbers to the next one; nothing in the later chapters is generic.

Bridge B33 33 m span, 6 girders	→	Analysis & prestressing grillage, IRC:6 / IRC:112	→	Healthy baseline commissioning tests, model updating, 1 year of data	→	Inspection, NDT, monitoring what, where, how often
Maintenance decision actions, intervals, reinspection	←	AI/ML assessment novelty, classification, regression, fusion	←	Features changes from own baseline	←	Synthetic data digital twin + 6 mechanisms

1.1 Assumed site and service conditions

The brief does not fix a site, so I assumed one that makes the later choices concrete. B33 carries a one-way, three-lane carriageway of a divided state highway (one of a pair of parallel bridges) over an irrigation canal and a service road in the coastal plain of West Bengal, about 120 km from the Bay of Bengal. It has three simply supported spans of 33 m on wall-type piers P1 and P2 and abutments A1 and A2; only one span length was selected, as required by §1.2, and span S2 between the piers is the one designed, instrumented and assessed in detail. The climate gives effective bridge temperatures of roughly 15 to 42 °C, heavy monsoon rain from June to September and wet-dry cycles the rest of the year; I therefore took the exposure as *severe* in the sense of IRC:112 Table 14.1, with 50 mm clear cover to links and 75 mm to ducts. The design life is 100 years. Traffic is mixed, with a high share of loaded three-axle and multi-axle lorries.

Outside the scope are the piers, abutments and foundations (they appear only in the inspection scheme), hydraulics and scour, seismic and wind design, and shop drawings of the prestressing system. All 'measurements' in the report are synthetic; how they were produced, and what that means for the results, is explained in chapters 10 and 11.

1.2 Codes and references used

Document	Used for
IRC:5-2015	carriageway, footpath and kerb widths
IRC:6-2017	dead loads, live loads (Class A, 70R), impact, footway load, lane reduction, temperature, load combinations (Annex B)
IRC:112-2020	materials, losses, stress limits, crack control, ULS flexure and shear, deflection
IRC:83 (Part II)-2018	laminated elastomeric bearings
IRC:SP:69-2011	expansion joints
IRC:SP:51-2015	load testing of bridges
IRC:SP:35-1990, IRC:SP:52-1999	inspection and maintenance practice, condition assessment
IRC:SP:40-2019, IRC:SP:80-2008	repair and strengthening techniques, corrosion monitoring and remedies
IS 516 (Part 5), ASTM C876	UPV and rebound hammer testing; half-cell potential

2 Bridge configuration

Cross-section. The carriageway is 11.0 m (three lanes of 3.5 m with 0.25 m marginal strips), bounded by 0.50 m New Jersey crash barriers. Outside each barrier there is a 1.75 m footpath, slightly wider than the IRC:5 minimum of 1.5 m because the footpath also carries service ducts for power and optical fibre, and a 0.25 m railing kerb. The overall deck width is therefore 16.0 m (Figure 1). The barrier between footpath and traffic is a deliberate choice: on a highway with fast heavy vehicles it protects pedestrians and it also keeps vehicles off the deck cantilever.

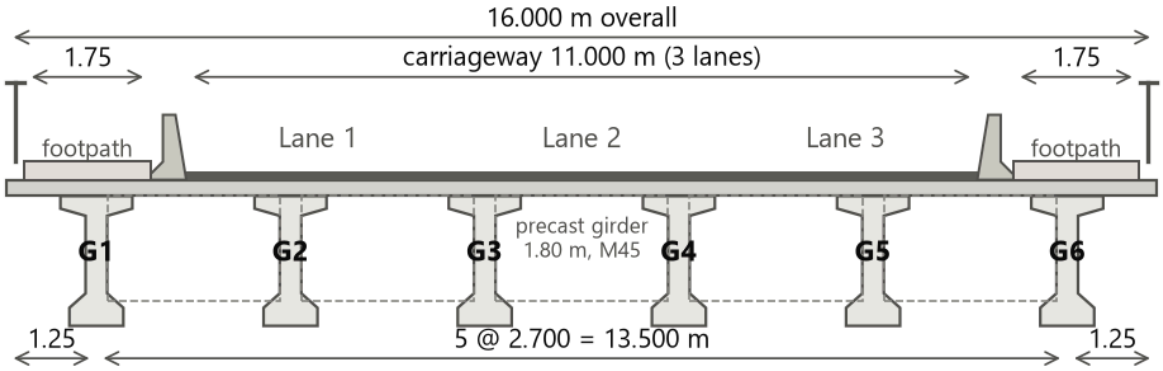


Figure 1. Cross-section of B33 at mid-span (dimensions in m): 225 mm M40 deck slab on six 1.80 m M45 precast girders; the mid-span cross girder behind the section is dashed.

Number and spacing of girders. The construction depth is limited: the deck has to clear the service road below and the approach embankments are already high, so I fixed the precast girder depth at 1.80 m (2.025 m with the slab, span/depth = 16.3). With the depth fixed, the girder number decides how much prestress each girder must carry. Table 3 compares four, five and six girders on a preliminary basis (Courbon distribution, the same section shape, the prestress needed for zero tension at the soffit under the characteristic combination).

Table 3. Girder-number study at 1.80 m depth (preliminary, Courbon; the 750 mm bulb takes at most four cables)

Girders	Spacing (m)	Slab (mm)	Courbon factor, outer	LL moment outer (kN m)	Pe needed (MN)	Strands per girder	12T15 cables	Girder mass (t)	Strands, deck	Concrete (m^3)
4	4.00	275	0.368	3618	9.67	63.5	6 - too many	83	254	279
5	3.25	250	0.297	2918	8.04	52.8	5 - too many	81	264	294
6	2.70	225	0.250	2457	6.78	44.5	4 - fits	77	267	304

Four girders would need about 64 strands each, i.e. six 12T15 cables, which cannot be placed in a 750 mm bottom bulb with the required duct spacing and cover, and the deck slab would have to span 3.0 m clear. Five girders need about 53 strands, five cables, still too many for the bulb at this depth. Six girders need about 45 strands, which four 12T15 cables (48 strands) provide with a small margin. Six girders use about 5 % more strand and 9 % more concrete than four, but they work within the depth limit, give a 225 mm slab with a short 1.7 m clear span and a 1.25 m cantilever, keep each girder under 80 t for lifting by two cranes, and leave more load paths if one girder is damaged. That last point matters for the later chapters: a deck with six girders sheds load from a weakened girder to its neighbours, which keeps the bridge safe but also makes damage harder to see in global measurements.

Girder section. The precast girder (Figure 2) has a 1000 x 175 mm top flange with a 100 mm taper, a 300 mm web and a 750 x 250 mm bottom bulb with a 200 mm taper. The web is 300 mm rather than the common 250 mm because a 92 mm HDPE duct has to pass through it with cover on both sides; the bulb width is set by three ducts side by side with clear spacing larger than one duct diameter. Over 1.5 m at each end the web is widened to the full bulb width (end block) and then tapers back over another 1.5 m, to take the four anchorages and the high shear near the bearings. Section properties are A = 0.855 m^2, y_b = 0.912 m, I = 0.3272 m^4 for the precast girder and I = 0.6711 / 0.6790 m^4 for the composite outer / inner girder (deck M40 transformed with m = 0.971). The full slab width is effective (IRC:112 cl. 7.6.1.2 gives more than the girder spacing for a 33 m span).

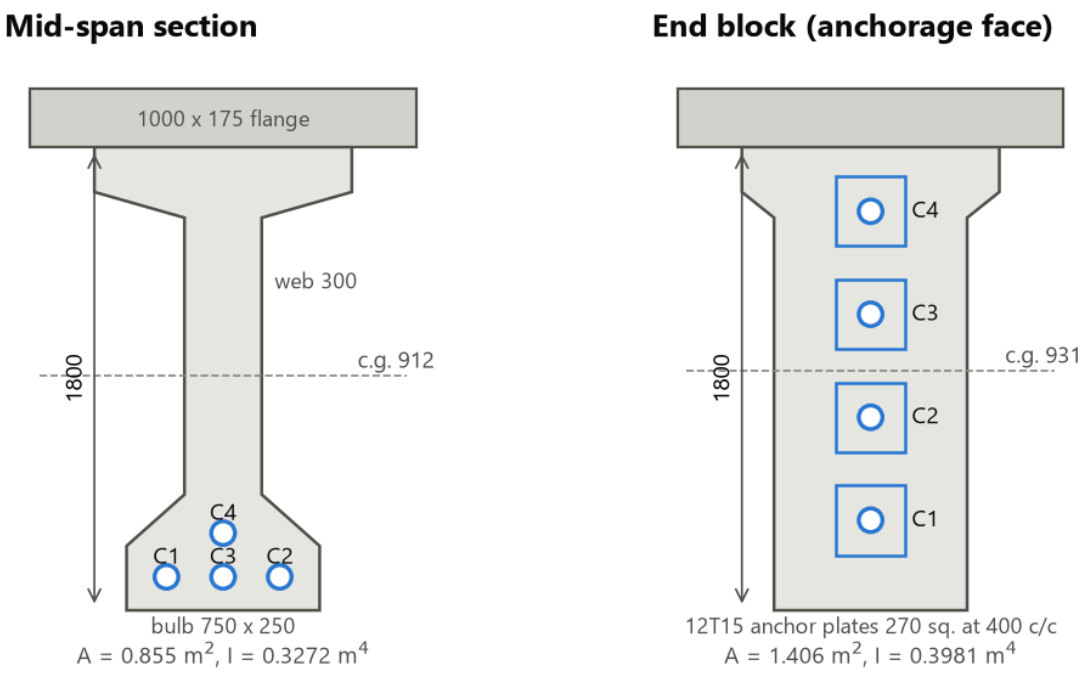


Figure 2. Precast girder at mid-span and at the anchorage face, with ducts and 12T15 anchor plates (mm).

Cross girders. End diaphragms (400 x 1600 mm) are needed to hold the girder ends, to transfer transverse loads to the bearings and, importantly for maintenance, as jacking beams for bearing replacement. For intermediate cross girders I compared one at mid-span with three at the quarter points in the grillage model: the extra two changed the outer girder live-load moment by less than 3 % and the first two natural frequencies by about 1 %, so one 300 x 1450 mm cross girder at mid-span was adopted. It also braces the girders laterally while the deck is cast.

Bearings and joints. Each girder sits on a 400 x 500 mm laminated elastomeric bearing at both ends (five 12 mm internal layers), so the span expands about its centre and both ends move. Strip-seal expansion joints (IRC:SP:69) are provided at each pier and abutment, with a 40 mm gap. The joint at a pier is the place where water reaches the bearings, the girder ends and the anchorages; this becomes the main thread of the year-18 assessment.

Materials. Girders M45 (stressing at 36 MPa, about 10 days), deck slab and cross girders M40, reinforcement Fe500D, strand 15.2 mm 7-wire low-relaxation (IS 14268 Class 2, f_{pk} = 1860 MPa, 140 mm^2), corrugated HDPE ducts. Unit weights from IRC:6 Table 1: 25 kN/m^3 for concrete and 22 kN/m^3 for the 75 mm bituminous wearing coat.

3 Structural analysis and preliminary design

3.1 Loads

Table 4. Loads on one span

Load	Value	Basis
Girder self-weight	21.4 kN/m (end block 35.2 kN/m)	on the precast girder
Deck slab (wet) and cross girder	15.2 kN/m per inner girder; 14.6 kN/m outer; cross girder 26.1 kN at mid-span	on the precast girders, tributary widths
Crash barrier, footpath, railing	7.3, 6.0, 1.5 kN/m each side	on the composite deck (grillage)
Wearing coat	1.65 kN/m^2 over 11.0 m	75 mm at 22 kN/m^3
Footway live load	2.78 kN/m^2	IRC:6 cl. 206.1, L > 30 m: P' = P - 260 + (4800/L)(16.5 - W)/15 with P = 400 kg/m^2, W = 1.75 m
Vehicles	Class A train; 70R wheeled (1000 kN) and tracked (700 kN)	IRC:6 Table 6A, three lanes: one 70R + one Class A, or three Class A, whichever is worse
Impact	Class A and 70R wheeled 4.5/(6+L) = 0.115; 70R tracked 0.10	IRC:6 cl. 208 (span > 12 m)
Multi-lane reduction	10 % for three lanes loaded	IRC:6 cl. 205
Temperature difference	positive 17.8 / 4.0 / 2.1 °C, reverse 10.6 / 0.7 / 0.8 / 6.6 °C	IRC:6 cl. 215, $\alpha = 12 \times 10^{-6}$/°C
Differential shrinkage	100×10^{-6} deck relative to girder, creep factor 0.43	eigenstresses only (simply supported)

Load combinations follow IRC:6 Annex B. For the ultimate limit state the factors are 1.35 on self-weight and SIDL, 1.75 on the wearing coat and 1.5 on traffic. For the serviceability checks I used the rare (characteristic) combination with live load 1.0 and temperature 0.6, the frequent combination with 0.75 and 0.5, and the quasi-permanent combination with live load zero and temperature 0.5.

3.2 Structural model

The deck was analysed with a grillage written for this work (Python, about 300 lines): six longitudinal members along the girder lines with the composite section properties, and transverse members every 1.375 m representing deck slab strips ($I = b\,t^3/12$, $J = b\,t^3/6$ following Hambly), replaced by T-beams at the cross girders. Each node has a vertical displacement and two rotations. Girder torsion constants are the sum of the rectangle values of flange, web and bulb plus half of the slab. Bearings are vertical springs of 1.6×10^6 kN/m. Loads between girder lines are shared to the four corner nodes, and loads outside the outer girders (barrier, footpath) are applied as a force and a torque on the outer line. For the live load, influence surfaces of moment and shear at 13 sections of each girder were calculated, and every vehicle was moved along the span in 0.25 m steps and across the carriageway in 0.05 m steps; the worst transverse arrangement was found by dynamic programming with the IRC kerb clearances (1.2 m for 70R, 0.15 m for Class A) and 1.2 m between vehicles. The self-weight of girders and wet slab acts on the precast girders individually and was calculated as for a simply supported beam with the 0.45 m overhangs.

The grillage was checked against hand results (Appendix B): with 10 kN/m on each girder the sum of mid-span moments is within 0.2 % of $wL^2/8$ and the inner girder deflection within 1.5 % of $5wL^4/384EI$ plus the bearing compression. For live load, Courbon's method is valid here (span/width = 2.06, stiff cross girders) and gives 2457 kN m for G1 against 2426 kN m from the grillage (Figure 3). For the inner girders the grillage is larger than Courbon, because a vehicle near the centre loads G3 and G4 more than a rigid-deck distribution assumes; I used the grillage values for design.

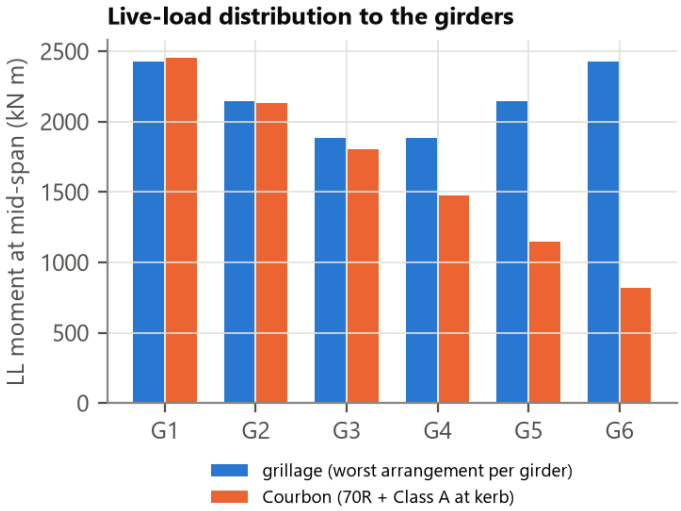


Figure 3. Mid-span live-load moment (with impact and lane reduction) of each girder: grillage against Courbon.

3.3 Load effects

Table 5. Mid-span moments per girder

Mid-span moment (kN m)	G1 (outer)	G2	G3
Girder self-weight	2930	2930	2930
Deck slab + cross girder	2097	2281	2281
Barrier, footpath, railing	628	716	667
Wearing coat	420	400	413
Vehicles + impact + footway (worst)	2719	2419	2098
ULS design moment M_{Ed}	12 446	12 329	11 805

For G1 the worst case is the 70R wheeled vehicle at the kerb clearance with a Class A train beside it; for the inner girders it is the same pair placed over them. The outer girder has the largest live-load moment, the inner girders carry more slab and the same SIDL share, and in the end the ULS moments of G1 and G2 are within 1 %; one girder design is used for all six. Envelopes for G1 are in Figure 4.



Figure 4. Bending moment and shear envelopes of girder G1 (characteristic components and ULS).

4 Prestressing system

4.1 Cables and profiles

Each girder has four cables of twelve 15.2 mm strands (12T15, A_p = 1680 mm² per cable) in 92 mm corrugated HDPE ducts, stressed from both ends to 0.75 f_{pk} (1395 MPa, 2344 kN per cable). The jacking stress is kept below both the IRC:112 maximum and the 0.765 f_{pk} used in older practice, to leave room for over-stressing if friction turns out higher than assumed on site.

The layout (Figures 2 and 5) came out of a clash check along the whole girder rather than from the mid-span section alone. At the anchorage face the four anchor plates are stacked in the web plane at 0.35, 0.75, 1.15 and 1.55 m above the soffit, 400 mm apart, which keeps the group centroid (0.95 m) at the end-block centroid (0.93 m). At mid-span three cables lie in the bottom row at 130 mm and one in a second row at 300 mm. The two outer cables of the bottom row must also move sideways by 220 mm, and a lateral swing is impossible inside the 300 mm web. My first layout tried to do it there. In the final layout these two cables drop into the bulb quickly (parabolas ending at 5 m and 8 m from the end) and swing sideways only once they are inside the bulb; the two centre cables follow full-length parabolas and never leave the web plane. The profile of each cable is a parabola from the anchorage with a horizontal tangent where it reaches its mid-span level, straight beyond that point, with a 12 mm allowance for the strands lying low in a draped duct.

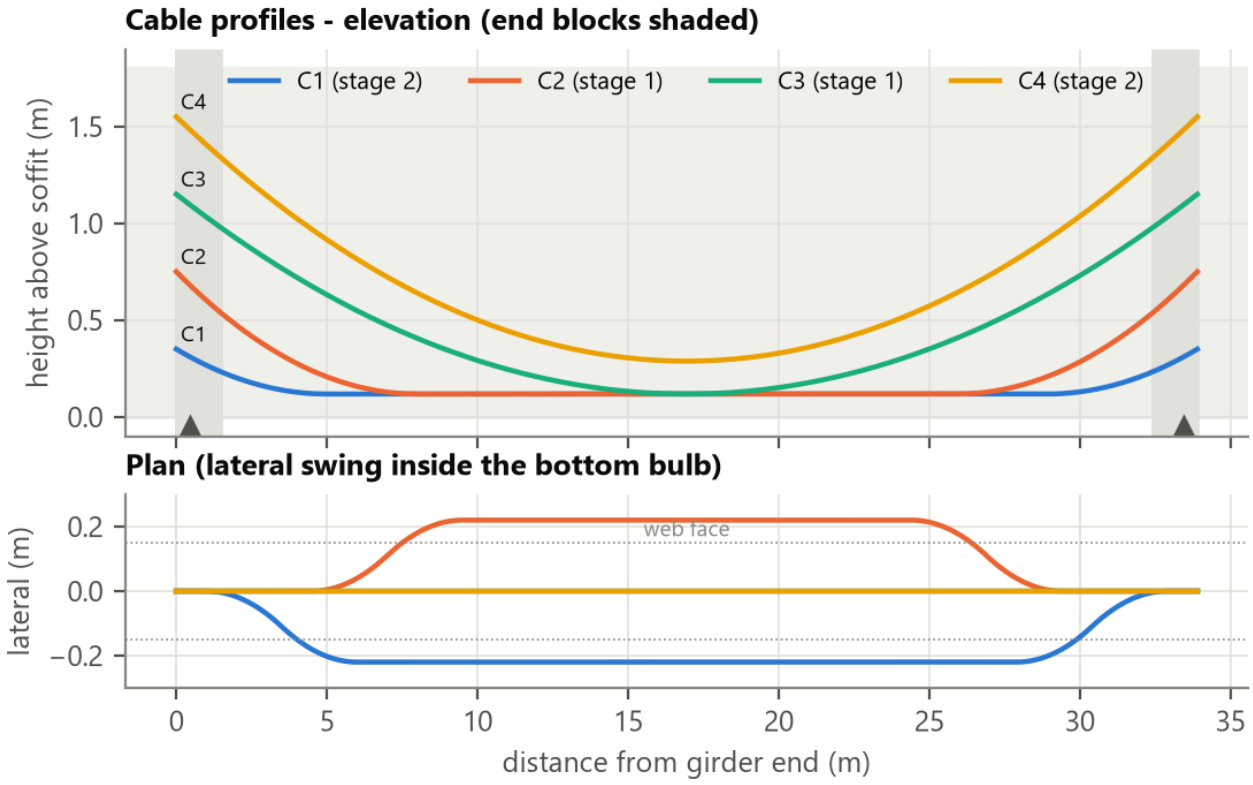


Figure 5. Cable profiles in elevation and plan. Cables C2 and C3 are stressed at 10 days, C1 and C4 at 28 days.

Two-stage stressing. Stressing all four cables at 10 days would give a soffit compression of about 18.6 MPa at mid-span against a limit of 0.48 f_{cj} = 17.3 MPa. I therefore stress C2 and C3 at 10 days (enough to lift the girder off the casting bed and carry its own weight) and C1 and C4 at 28 days. Both groups have their centroid at the end-block centroid, so the girder ends are not bent at either stage.

4.2 Losses and effective prestress

Friction follows $P(x) = P_j\, e^{-(\mu\theta + kx)}$ with μ = 0.17 and k = 0.002/m for HDPE ducts, θ being the cumulative vertical and plan angle from the nearer anchorage. Draw-in of 6 mm was handled by the area method. Elastic shortening was calculated cable by cable in the stressing order C2, C3, C1, C4, using the concrete modulus at the age of each stressing. Creep, shrinkage and relaxation follow the time-dependent loss expression of IRC:112 (same form as EN 1992 eq. 5.46):

$$\Delta\sigma_{p,c+s+r} = [\, \varepsilon_{cs}E_p + 0.8\Delta\sigma_{pr} + (E_p/E_{cm})\,\varphi(t,t_0)\,\sigma_{c,QP}\,] \,/\, [\, 1 + (E_p/E_{cm})(A_p/A_c)(1 + A_c z_{cp}^{\,2}/I_c)(1 + 0.8\varphi(t,t_0))\,] \qquad (1)$$

with creep and shrinkage from IRC:112 Annex A2 (RH 70 %, notional size 281 mm, cylinder strength taken as 0.8 of the cube strength), the quasi-permanent concrete stress at the tendon level (10.8 MPa at mid-span) and final relaxation taken as three times the 1000-hour value for the actual initial stress ratio (IRC:112 cl. 6.3.6). Table 6 gives the results per cable at mid-span.

Table 6. Prestressing forces and losses at mid-span (one girder)

Cable	Stressed at (d)	P_j (kN)	Friction + draw-in (kN)	Elastic shortening (kN)	P_0 (kN)	$\varphi(\infty,t_0)$	Relaxation (MPa)	c+s+r loss (MPa)	P_e (kN)
C1	28	2344	179	57	2107	1.63	80	170	1821
C2	10	2344	202	191	1951	1.99	47	168	1668
C3	10	2344	124	122	2098	1.99	78	187	1784
C4	28	2344	134	0	2210	1.63	104	186	1898
Total		9374	639	370	8366				**7172**

The immediate loss at mid-span is 10.8 % and the long-term loss a further 12.7 %, giving **P_0 = 8.37 MN and P_e = 7.17 MN, P_e/P_j = 0.765**. These numbers are of the expected order for a 34 m post-tensioned girder (total loss in the low twenties of per cent), which I took as a first sanity check. C2 loses most to elastic shortening because three cables are stressed after it. Figure 6 shows the force along the girder: the draw-in of the straight-ended cables reaches nearly to mid-span because friction there is only wobble.

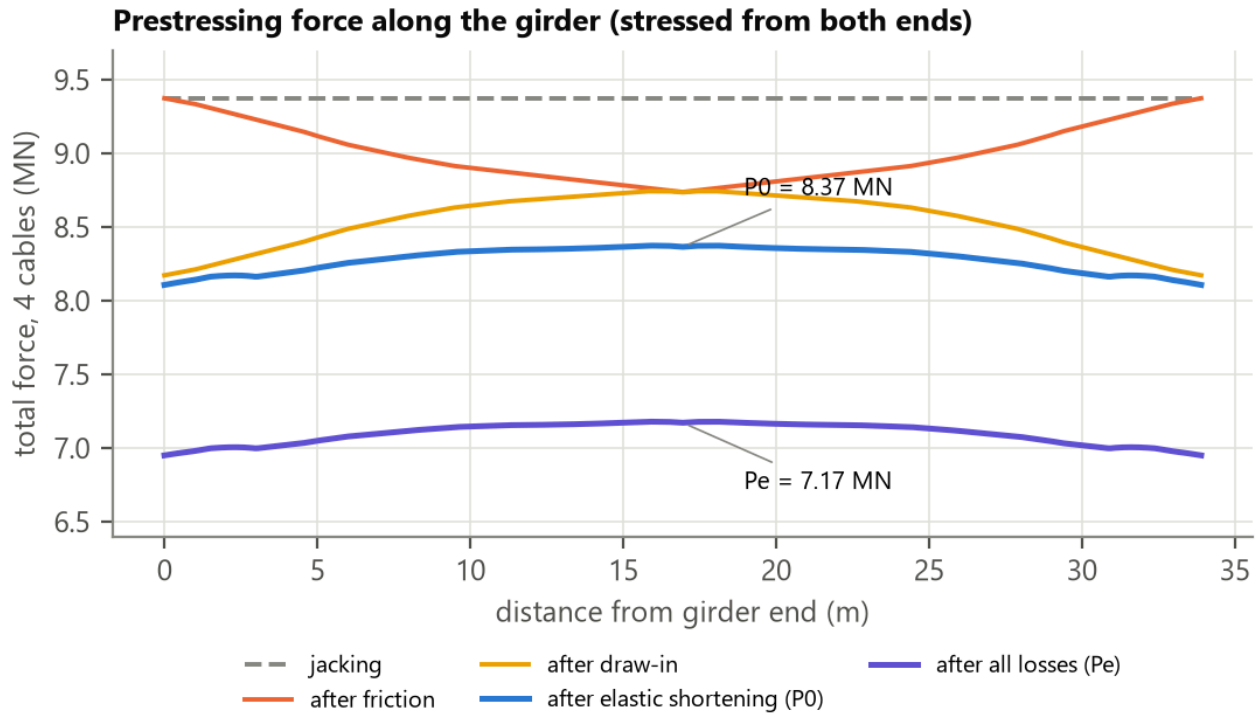


Figure 6. Total prestressing force of one girder along its length at the successive stages.

4.3 Stresses at the important stages

Stresses were checked at mid-span, L/4 and two sections near the support for every girder, at five stages: transfer of the first two cables (10 d, self-weight only), all cables stressed (28 d), deck slab cast (60 d), opening to traffic (90 d) and long-term service (t = ∞) in the three combinations. For the service checks the characteristic prestress was used, 0.9 P_m where prestress is favourable and 1.1 P_m where it is not (IRC:112, post-tensioning), and the temperature difference and differential shrinkage were added only where they are adverse. My design targets, which are stricter than the crack-width requirement of IRC:112 Table 12.1 for bonded tendons, were: no decompression of the soffit under the quasi-permanent combination, soffit tension below f_{ctm} (uncracked section) under the rare combination, compression below 0.48 f_{ck} (rare) and 0.36 f_{ck} (quasi-permanent).

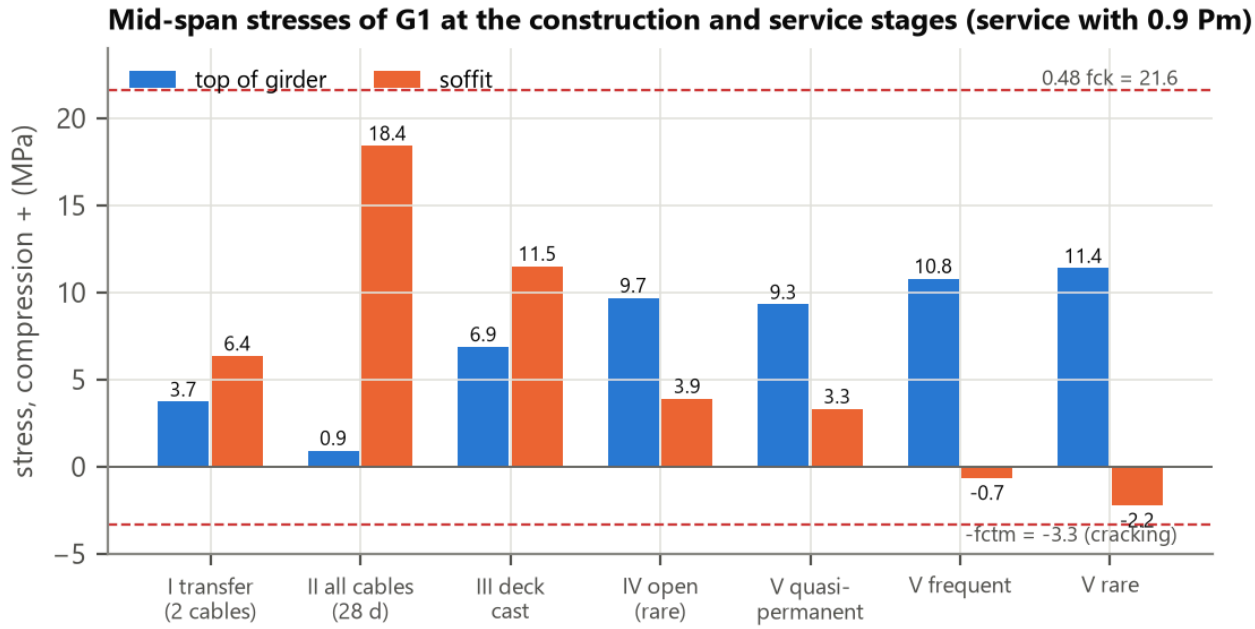


Figure 7. Mid-span stresses of the outer girder G1 at all stages.

Table 7. Stress checks (compression positive)

Check (worst of G1-G3 and all sections)	Value (MPa)	Limit (MPa)	Status
Transfer, stage 1: max compression / min stress	6.79 / +2.39	17.28 / ≥ 0	OK
All cables stressed: soffit compression	18.42	21.60	OK
All cables stressed: top fibre	+0.92	≥ 0 (no tension)	OK
Long term, quasi-permanent, soffit (0.9 P_m)	+2.68	≥ 0 (decompression)	OK
Long term, frequent, soffit (0.9 P_m)	-0.85	≥ -3.3 (uncracked)	OK
Long term, rare, soffit (0.9 P_m)	-2.27	≥ -3.3 (uncracked)	OK
Rare, top of girder / top of slab (1.1 P_m)	10.40 / 6.48	21.6 / 19.2	OK
Quasi-permanent, soffit compression (1.1 P_m)	8.76	16.2	OK

The governing check is the soffit under the rare combination at mid-span (-2.27 MPa). The reverse temperature difference alone contributes about 1.5 MPa of tension there, so the design would not satisfy a 'no tension under rare loads' rule, but the section stays uncracked with a margin of about 1 MPa. I kept this margin deliberately visible because it decides, later, how much prestress the girder can lose before it cracks.

4.4 Ultimate limit state, shear, deflection and bearings

Flexure. Strain compatibility at mid-span (rectangular block 0.8x at $f_{cd} = 0.67\ f_{ck}/1.5$, strands bilinear with $f_{pd} = f_{p0.1k}/1.15$, pre-strain from P_e) gives a neutral axis depth of 252 mm (block depth 201 mm, inside the 225 mm slab), strand strain 28 x 10^{-3} (yielded, ductile failure, x/d = 0.14) and M_{Rd} = 16 468 kN m against M_{Ed} = 12 446 kN m for G1, a utilisation of 0.76. The untensioned bottom bars were ignored, which is conservative.

Shear. The critical section is where the web has just narrowed to 300 mm, 2.75 m from the bearing. With the duct in the web the nominal width is b_w - 1.2φ = 190 mm (HDPE ducts). For the worst girder V_{Ed} = 1452 kN, the vertical component of prestress (0.9 P_e sinα) is 532 kN, and the net 921 kN is below $V_{Rd,max}$ = 1329 kN for cotθ = 2. The required links are 771 mm²/m; I adopted 2-legged 12 mm links at 200 mm (1131 mm²/m) over the end 6 m and 2-legged 10 mm at 250 mm elsewhere (above the minimum of 183 mm²/m).

Deflection and camber. The live-load deflection with impact is 11.8 mm for G1 (span/2786), well within span/800 = 41.2 mm. The camber is 37 mm upward when all cables are stressed and about 11 mm after the deck is cast. The elastic camber due to P_e alone is 57 mm; this number reappears in chapter 10, because a loss of prestress shows up as a loss of this camber.

Bearings. The largest reaction is 1222 kN (inner girder, permanent load 794 kN). For the 400 x 500 mm bearing with five 12 mm layers the shape factor is 9.0 and the total design strain from compression, shear (movement 14 mm from creep, shrinkage and temperature) and rotation is 1.84, below 7 (IRC:83 Part II); the minimum pressure 4.2 MPa is above 3 MPa, so the bearing will not walk. In service each bearing shears by about 0.192 mm per °C (αL/2), which is the quantity the bearing transducers will watch.

Anchorage zone and other details. The end block was sized to the anchor plates (spacing 400 mm, edge distance 250 mm at the top) and to the bursting forces of the local zones; the final anchorage reinforcement has to follow the approval document of the prestressing system actually used. The deck slab was checked separately for 70R wheel loads on its 1.7 m clear span; these local designs are not repeated here because they do not feed the later chapters.

5 Healthy baseline

A healthy baseline is not the design calculation. The as-built bridge is always somewhat different: concrete is stiffer than the code modulus, barriers and footpaths stiffen the deck, bearings are not ideal. What later measurements must be compared with is the bridge as it was measured when it was new, together with a model that reproduces those measurements. I set the baseline up in three steps.

5.1 Commissioning tests

Before opening, a 30-minute ambient vibration test was recorded at night with all 12 accelerometers, and a static load test was carried out with four 300 kN three-axle lorries (two abreast, two in tandem, centred at mid-span) in the spirit of IRC:SP:51. In this synthetic study the 'real' bridge is a model whose stiffness parameters are hidden from the assessment (concrete modulus 6 % above E_{cm}, crash barriers acting at about one third of their stiffness, dynamic modulus 24 % above static). The commissioning record was identified without using any model shapes, by frequency domain decomposition of the 12 channels (Figure 8): peaks of the first singular value were picked, each peak was named after the design-model mode its singular vector resembled most, and its frequency and damping were refined by fitting a single-degree-of-freedom spectrum around the peak.

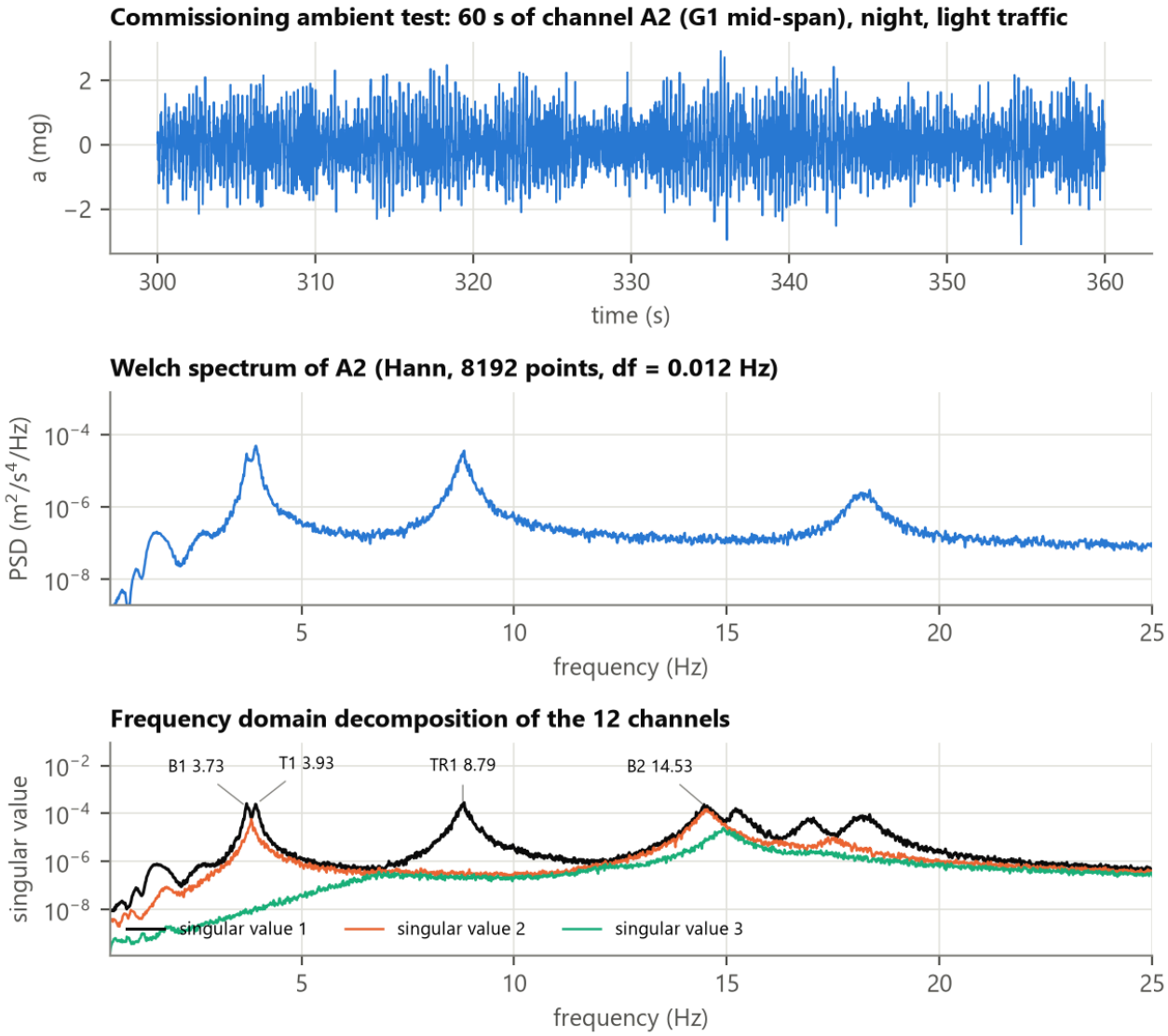


Figure 8. Commissioning ambient test: one channel in time and frequency, and the FDD singular values of all 12 channels.

I first identified the frequencies by filtering the channels with the design-model mode shapes, which is convenient but wrong at this stage: with the first bending and torsion modes only 5 % apart, the model shapes leaked one mode into the other and gave f_2 = 3.78 Hz, while the model-free FDD gave 3.93 Hz. The lesson is that before the model has been checked against the structure, the data have to speak for themselves. Table 8 compares the design model with the measurements.

5.2 Model updating

Three parameters were updated by weighted least squares on the four identified frequencies and the six girder deflections of the load test: a factor on the concrete modulus (E_{scale}), the share of crash-barrier stiffness acting with the deck (β) and the ratio of dynamic to static modulus (E_{dyn}). Frequencies were weighted with a standard error of 0.2 % and deflections with 0.10 mm.

Table 8. Design model, commissioning measurements and the updated twin

Quantity	Design model	Measured (year 0)	Updated twin
f_1 bending (Hz)	3.466	3.730	3.731
f_2 torsion (Hz)	3.562	3.926	3.922
f_3 transverse (Hz)	8.345	8.794	8.793
f_4 2nd bending (Hz)	13.168	14.530	14.540
Load-test deflection G1 / G3 (mm)	4.99 / 6.90	4.36 / 6.32	4.40 / 6.34
E_{scale}, β, E_{dyn}	1.00, 0, 1.20	-	1.055 ± 0.008, 0.36 ± 0.02, 1.245 ± 0.010

The design model is 7 to 10 % too soft in frequency and 9 to 14 % too flexible in the load test, which is normal for a design model. After updating, the twin reproduces all ten measurements within 0.2 % in frequency and 0.15 mm in deflection, and the measured and twin mode shapes agree with MAC above 0.99 for the first three modes. The updated parameters (1.055, 0.36, 1.245) are close to the hidden values (1.06, 0.35, 1.24), which is reassuring, but I treat the twin as a calibrated surrogate rather than the truth: E_{scale} and β are correlated (correlation -0.29) and a different pair could fit nearly as well. The second bending mode (14.5 Hz) is almost repeated with the second torsion mode, its measured shape is a mixture of the two (MAC 0.47), and for that mode the twin shape, not the measured one, is used in the spatial filter later.

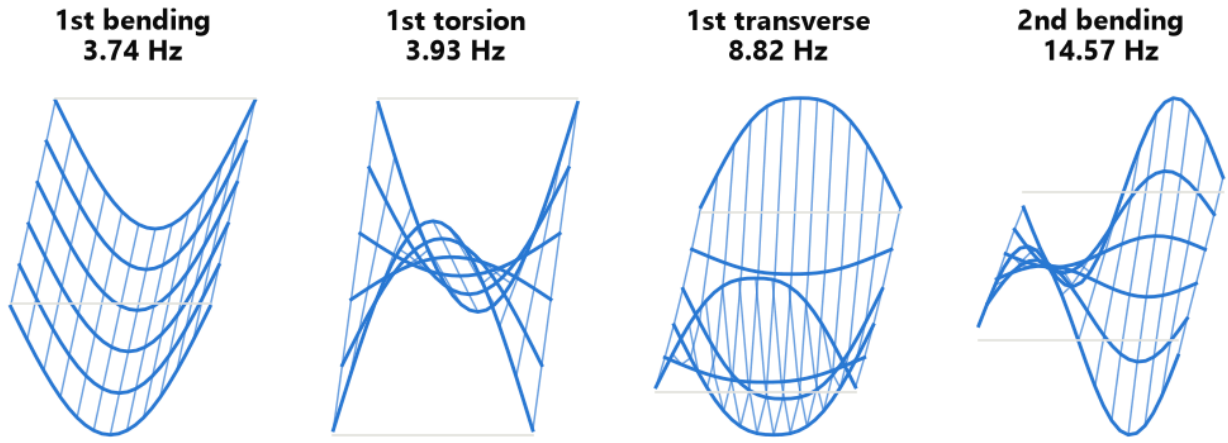


Figure 9. First bending, torsion and transverse modes and the second bending mode of the updated twin (25 °C).

5.3 One year of healthy monitoring

The permanent system then recorded two 30-minute records a day for a year (02:00 and 14:00), 730 records in all. The first bending frequency moved between 3.65 and 3.79 Hz over the year, a range of almost 4 %, driven by temperature (-0.149 %/°C, from the concrete modulus), by the traffic present during the record and by the extra mass of a wet deck in the monsoon. A linear model in temperature and $\log(a_{RMS})$, the RMS acceleration being a measure of how busy the bridge was, was fitted for each frequency (Figure 10). After normalisation the scatter of f_1 is 7.0 mHz (0.19 %), and the remaining structure in the residuals is the monsoon mass effect, which I left in on purpose because on a real bridge its size is not known in advance.

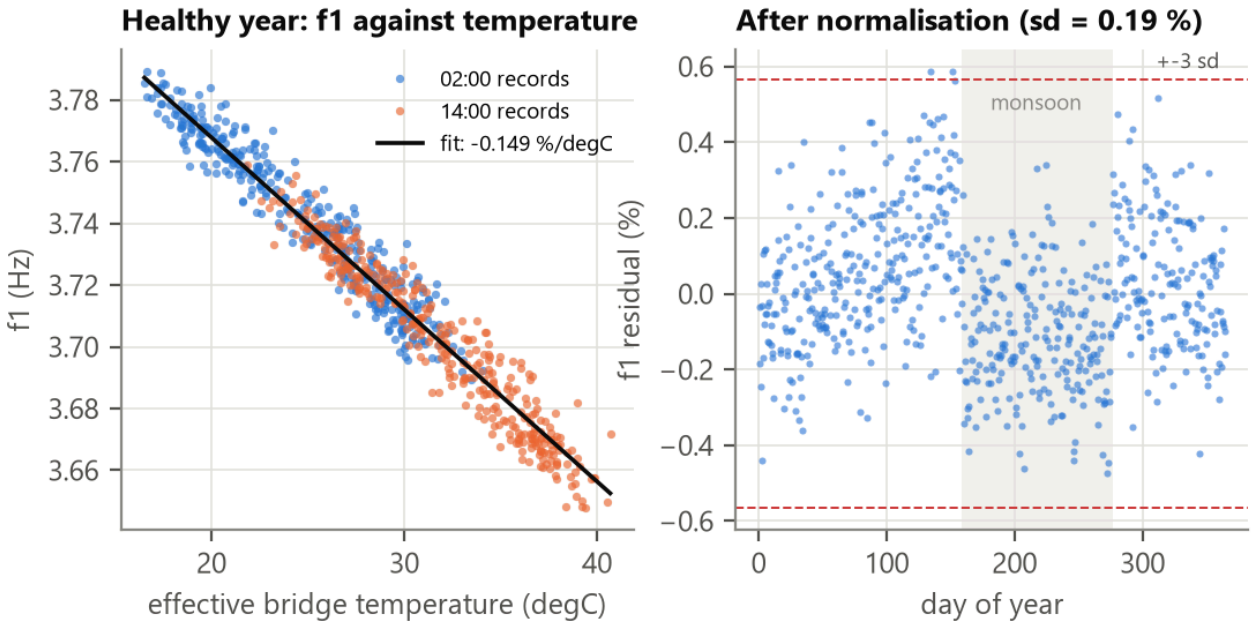


Figure 10. Healthy year: f_1 against temperature, and the residual after temperature/traffic normalisation.

Table 9. Baseline structural and monitoring data of the healthy bridge

Baseline quantity (span S2, healthy, 25 °C reference)	Value
Effective prestress per girder, mid-span	P_e = 7.17 MN (design), soffit precompression from P_e 23.4 MPa
Soffit stress, quasi-permanent (mean prestress)	about 5.7 MPa compression (G1)
Load test, 4 x 300 kN: mid-span deflection G1..G3	4.38 / 5.54 / 6.31 mm
Load test: soffit strain G1..G3	60 / 70 / 66 microstrain
Natural frequencies f_1..f_4	3.741 / 3.932 / 8.815 / 14.572 Hz
Temperature sensitivity of f_1..f_4	-0.15 / -0.14 / -0.14 / -0.13 %/°C
Residual scatter after normalisation	0.19 / 0.17 / 0.16 / 0.16 %
Ambient acceleration, mid-span	0.6-0.8 mg RMS at night, 1.3-1.5 mg RMS in daytime traffic, peaks 10-25 mg
Bearing movement	0.192 mm/°C at both ends
NDT (girders, year 1, mean ± s.d.)	rebound 42.0 ± 2.2; UPV 4.56 ± 0.07 km/s; cover 53 ± 5 mm; HCP -133 ± 30 mV; resistivity 50 ± 10 kΩcm

Everything measured later is compared with this table and with the normalisation models, not with the design calculation. This 'own baseline' idea is also what lets a model trained on simulated data be applied to the real bridge (chapter 12).

6 Visual inspection scheme

Visual inspection is the cheapest source of information and, for most defects that matter on a bridge like this (water, corrosion, spalling, bearing and joint problems), the first. The scheme is organised as the brief asks, Bridge → Span → Component → Location → Observation, so that every record has an address that can be found again at the next inspection and compared.

Table 10. Inspection hierarchy and items

Component	Locations recorded	What to look for
PSC girders G1-G6	zones Z1-Z5 along the span (Z1/Z5 end zones 0-3.3 m with anchorages and bearings, Z2/Z4 quarter spans, Z3 mid-span); soffit, web faces, anchorage faces	flexural cracks at the soffit (any crack is significant in a fully prestressed girder), longitudinal cracks along ducts, rust staining, spalling, honeycombing, efflorescence, impact scrapes over the service road
Deck slab	bays between girders, cantilevers, construction joints	cracking, leakage and efflorescence from above, spalling at drains
Cross girders / diaphragms	end diaphragm A and B, cross girder L/2	cracking near jacking points, leakage from the joint
Bearings	G1-A ... G6-B	bulging, splitting, excessive or no shear deformation, debris, corrosion of plates, loss of contact
Expansion joints	EJ-A1, EJ-P1, EJ-P2, EJ-A2	torn or missing seal, debris in the gap, water running through, noise
Drainage	spouts U-1..U-6, D-1..D-6 per span	blockage, spouts too short (water on the girder), broken pipes
Footpaths, crash barriers, wearing coat	upstream / downstream, lanes	loose cover slabs, barrier cracking, ravelling and potholes near joints
Pier caps, piers, abutments	P1, P2, A1, A2	leakage stains, pedestal spalling, cracking, vegetation, scour signs

Condition states. Each observation is given a condition state from 1 (good) to 5 (critical), with written rules so that two inspectors reach similar ratings. For cracks the width bands are stricter on prestressed girders (CS2 above 0.05 mm, CS3 above 0.10, CS4 above 0.20, CS5 above 0.30 mm) than on reinforced elements (0.10 / 0.20 / 0.30 / 1.0 mm): a flexural crack in a girder designed to stay uncracked means its precompression has gone. Spalling is CS3, or CS4 with exposed steel, and CS5 with exposed strands over a large area. A component takes the worst state of its observations, and a bridge condition index from 0 to 100 is the weighted average of (5 - CS)/4 over the component groups, with weights reflecting their structural importance (girders 0.30, deck 0.12, bearings 0.12, joints 0.08, piers and pier caps 0.08 each, and so on).

Types and frequency of inspection. Routine inspections, a walk-over with binoculars and the under-bridge view from the service road, are carried out before and after every monsoon, because most defects here are water-driven. Principal inspections, close visual inspection of every element with an under-bridge unit and hammer tapping of suspect concrete, are made every three years and after any special event (overloaded vehicle, flood, vehicle impact). A special inspection follows any monitoring alarm. Inspection data are kept in the format above so that they can be used directly as features (chapter 12).

Digital inspection application. I wrote a single-file web application for the scheme (`inspection_app/index.html`, runs offline in any browser). It walks the inspector through span, component, location and observation, suggests the condition state from the crack width or defect type (the inspector can override it), stores photos, keeps the records in the browser, and exports CSV/JSON. It also draws the worst condition state of every girder zone as a map and computes the component ratings and the bridge condition index. The year-18 inspection of chapter 14 is built in as an example.

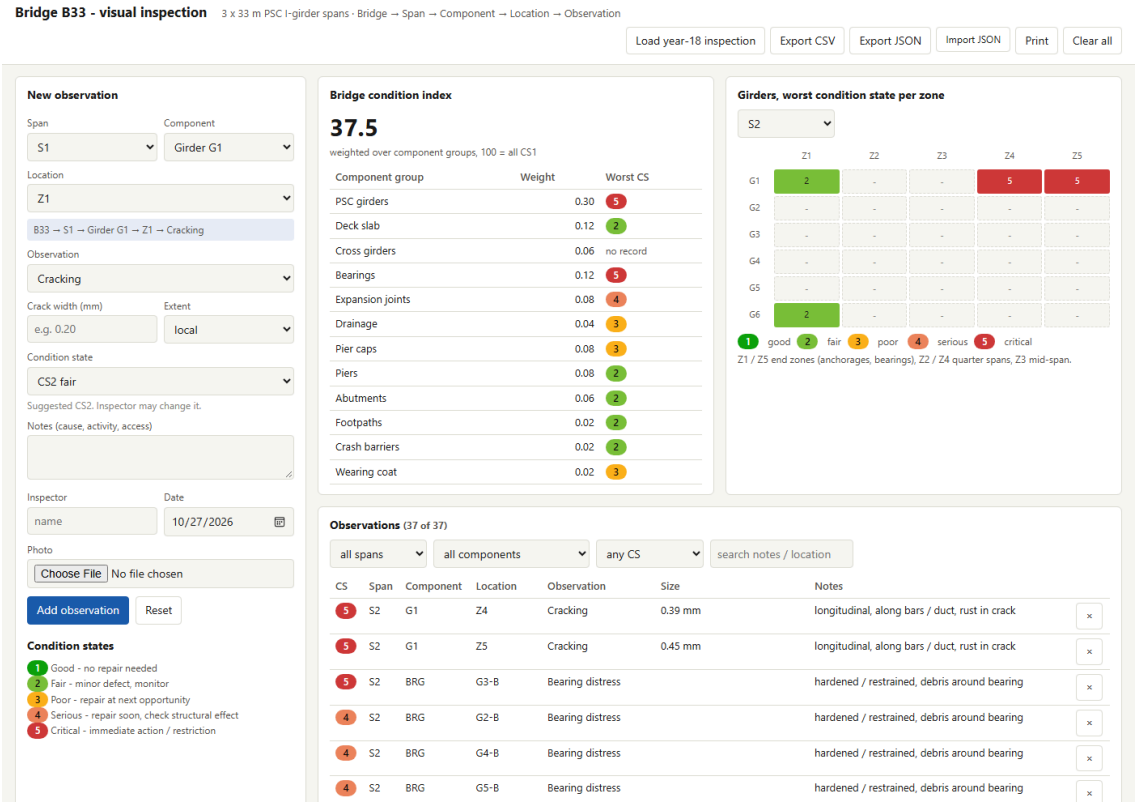


Figure 11. The inspection application with the year-18 inspection loaded: entry form, component ratings and bridge condition index, worst condition state per girder zone of span S2, and the observation list.

7 Non-destructive testing programme

An NDT method is chosen to answer a question raised by the inspection, not because it is available. The table below is the programme for B33 written in the order of the brief: observed defect, engineering question, test, measurement, interpretation.

Table 11. Defect - question - NDT - measurement - interpretation

Observed defect	Engineering question	NDT	Measurement	Interpretation
Rust staining, spalling, damp patches on girder soffit	Is the reinforcement corroding, and over what area?	Half-cell potential (ASTM C876) on a 0.5 m grid; Wenner resistivity	potential (mV vs CSE), resistivity (kΩcm)	< -350 mV: corrosion likely; -200 to -350: uncertain; > -200: low risk. Resistivity < 10 kΩcm means corrosion can proceed fast. A wet surface also shifts potentials negative, so both are read together
Same, and low-cover zones	Has carbonation reached the steel? Is the cover as specified?	Cover meter; phenolphthalein on drill-holes or small cores	cover (mm), carbonation depth (mm)	carbonation ≥ cover in wet concrete: depassivated, corrosion has started or will start
Cracks, honeycombing, surface deterioration	Is the concrete sound through the section? Where are the voids?	Ultrasonic pulse velocity (IS 516 Part 5/Sec 1), direct across web and bulb	pulse velocity (km/s)	> 4.40 good, 3.75-4.40 medium, < 3.75 doubtful (M25 and above); local drop over a crack = crack depth by indirect method
Same	Is the surface concrete weaker?	Rebound hammer (IS 516 Part 5/Sec 4) on a grid	rebound number	comparative only; corrected for carbonation, which hardens the skin; calibrated against a few cores
Rust or leakage at anchorages, longitudinal cracks along ducts	Are the tendons grouted? Are strands corroding inside the duct?	GPR to locate ducts, impact-echo along the duct line, borescope through a small drilled port	duct positions, void indications, visual of strands	voids at high points and anchorages are the usual sites of strand corrosion; HCP does not see strands inside ducts
Sag of the girder, cracks at mid-span	How much prestress is left?	Lift-off test at an accessible anchorage, or saw-cut stress-release on the soffit with strain gauges	force at lift-off (kN), released strain (microstrain)	direct measure of the remaining force, the only way to confirm what the monitoring suggests
Bearing distress	Is the bearing still free to move and rotate?	Bearing displacement transducers vs temperature; visual and dimension check	movement per °C (mm/°C)	much less than αL/2: restrained

Test locations. Each girder has five test areas, one per zone, on the bottom bulb and on the web (both sides on the outer girders), so a span has 30 areas and the bridge 90; the same areas are used at every inspection, which makes trends possible. Extra half-cell grids and carbonation tests are placed at the girder ends next to the joints, under every drainage spout of the outer girders, and wherever the inspection finds rust or spalling. Tendon investigations are limited to the anchorage zones and to duct high points, where voids collect. In this study the 30 areas of S2 enter the learning data as span-level summaries (lowest UPV, most negative potential, and so on) and the 90 areas of the whole bridge are clustered in chapter 13.

8 Structural response measurement

Three response quantities are monitored, each because it answers a question the others cannot.

Table 12. Monitoring quantities, sensors and their use

Quantity	Sensor and location	Sampling and duration	Interpretation
Displacement (load test)	LVDTs on a reference frame or a total station with prisms, mid-span of all six girders	static readings at each load step; test once at commissioning and at every principal / special inspection	deflection against the baseline, corrected to 25 °C; distribution across girders shows which girder has softened
Geometry (camber)	precise levelling points on the kerbs over G1 and G6 and at the deck centreline, mid-span and supports	once a year, same season and time of day	loss of camber beyond the creep prediction is the clearest sign of prestress loss (about 1 mm per 1 % loss here)
Strain, long term	vibrating-wire gauges cast into the bottom bulb at mid-span of G1, G3, G4, G6, with thermistors	hourly, permanent; daily averages used	drift towards tension beyond the predicted creep and shrinkage means precompression is being lost (about 12 microstrain per 1 %)
Strain, load test	electrical resistance or FBG gauges on the soffit at mid-span	100 Hz during the load test	local stiffness change, crack opening if strain jumps
Bearing movement	displacement transducers across the bearings of G1 and G6 at both ends	every 10 min, permanent	slope of movement against temperature, healthy 0.192 mm/°C; a drop at one end with a rise at the other is a restrained bearing
Temperature	thermistors in deck slab, web and bulb at mid-span of G1 and G4	every 10 min	effective temperature and gradient for normalising all other data
Acceleration	12 vertical accelerometers (chapter 9)	100 Hz, 30 min records twice a day	frequencies, mode shapes, damping, traffic response

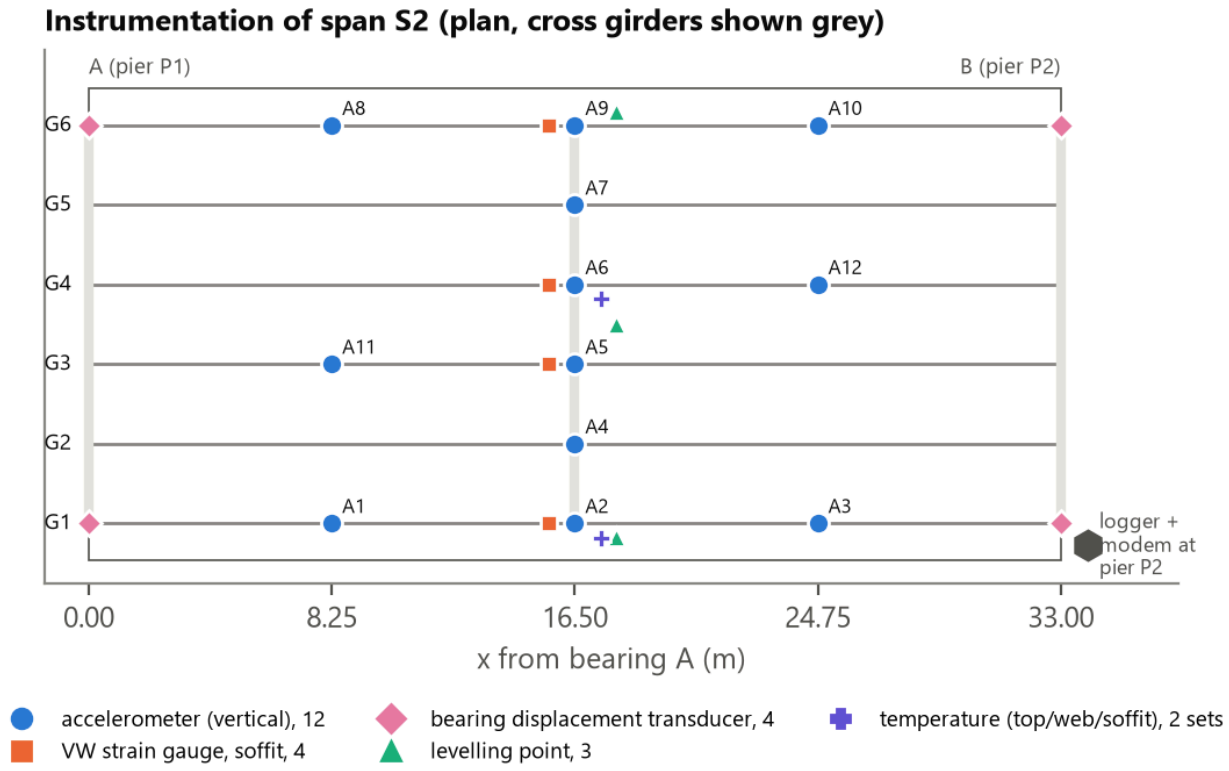


Figure 12. Instrumentation of span S2. The logger and a cellular modem sit in a cabinet on pier P2.

9 Vibration measurement and signal processing

Accelerometers. Twelve uniaxial vertical accelerometers are fixed to the soffits of the bottom bulbs (Figure 12): at L/4, L/2 and 3L/4 on both outer girders, at L/2 on the four inner girders, and at L/4 of G3 and 3L/4 of G4. They are low-noise force-balance or piezoelectric seismic sensors (noise about 2 µg/√Hz including cable and digitiser, range ±0.5 g), because night-time ambient response is only 0.5-1 mg RMS and MEMS sensors with 20-25 µg/√Hz would bury the higher modes in noise. Vertical direction only: the modes of interest are vertical bending and torsion; lateral and longitudinal motion of a 33 m span on elastomeric bearings carries little information on the girders.

The layout was not right first time. With the first ten sensors (outer girders at three points, inner girders at mid-span) the AutoMAC matrix of the twin's mode shapes showed two pairs of modes that were indistinguishable at the sensors, MAC = 1.00 (Figure 13): the second bending and second transverse modes both have zero response at mid-span and the same signs at the outer-girder quarter points; the second torsion mode and a higher transverse mode likewise. The shape matrix was singular (condition number of order 10^{12}) and modal filtering impossible. Two sensors at the quarter points of the inner girders G3 and G4 reduce the worst off-diagonal MAC to 0.37 and the condition number to 3.1.

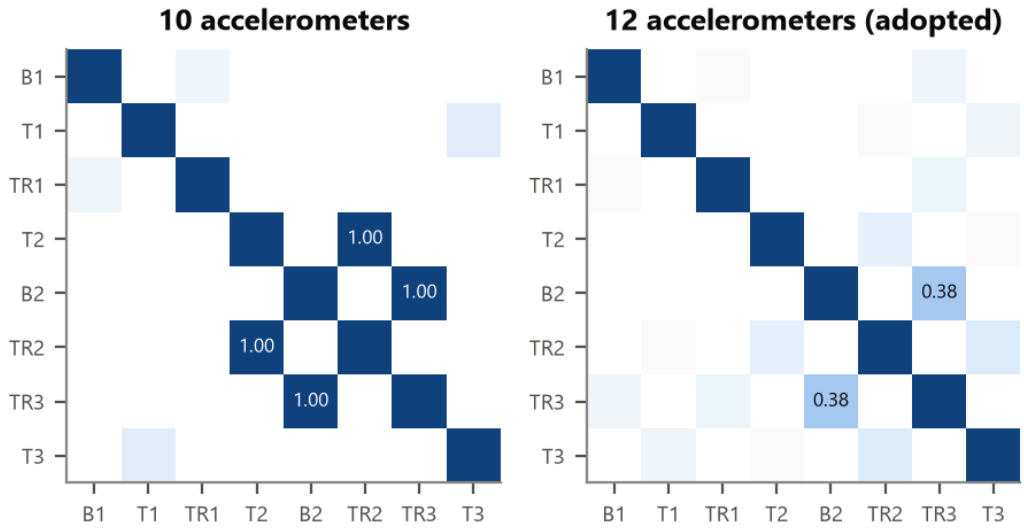


Figure 13. AutoMAC of the first eight twin modes at the sensor positions, before and after adding two quarter-span sensors.

Sampling and duration. The loggers sample at 1 kHz with a hardware anti-alias filter and decimate to 100 Hz, so that modes up to about 20 Hz are well inside the 40 Hz usable band. A record lasts 30 minutes, more than 6000 cycles of the first mode; with 10-minute records the scatter of f_1 in my simulations was about ±0.5 %, too large next to the effects to be detected, and 30 minutes with a fitted spectrum brought it to about 0.1 %. Records are taken at 02:00, when traffic is light and the excitation is closest to broadband, and at 14:00 for the traffic response.

Excitation. Ambient excitation is wind, ground micro-tremor and above all traffic. Traffic is not white noise: lorries bounce on their suspensions at about 1.3-2.6 Hz and pass at 40-60 km/h, and their mass adds to the bridge while they are on it. These effects bias the identified frequencies by up to ±0.2 % depending on how busy the bridge is, which is why a_{RMS} enters the normalisation. A drop-weight or shaker test is not needed for routine monitoring but would be used to calibrate damping or mode shapes in a special investigation.

Signal processing. The chain applied to every record is the one in the brief, with two additions:

- a(t): 12 channels, 30 min at 100 Hz; mean removed, 4th-order Butterworth band-pass 0.5-30 Hz (zero phase);
- FFT: Welch averaging, Hann window of 8192 points (82 s, Δf = 0.012 Hz), 50 % overlap, about 43 averages;
- frequency spectrum: auto-spectra of each channel and the full 12 x 12 cross-spectral matrix;
- modal filter: the channels are combined with the pseudo-inverse of the baseline mode-shape matrix, which separates the closely spaced bending and torsion modes;
- natural frequencies: a single-degree-of-freedom spectrum is fitted around each peak of the filtered spectrum (frequency and damping), and the mode shape is the singular vector of the cross-spectral matrix at the peak (FDD), compared with the baseline shape by MAC.

Damping estimates from ambient data are poor (scatter of ±25 % in my simulations) and were kept as a feature only to show that they are not useful here.

10 Deterioration scenarios

The brief leaves the type, location and size of deterioration to the student. I chose six mechanisms that are realistic for a post-tensioned girder bridge in a wet coastal climate, gave each a cause on *this* bridge, a range of values that I could defend, and a physical model of how it changes the structure. The ranges are deliberately wide, from barely detectable to clearly serious, because the AI/ML part has to learn where detection starts.

Table 13. Deterioration mechanisms and their models

Mechanism	Cause on B33	How it is modelled in the twin	Range and reason
Loss of prestress (PL)	creep, shrinkage and relaxation larger than predicted; grout voids and strand corrosion near anchorages; anchorage slip	ratio η of actual to expected P_e for one girder (60 %) or all girders (40 %). No change of uncracked stiffness (bonded tendons). Camber and VW strain change with creep factor $1 + \chi\varphi = 1.8$. Cracking when the soffit stress under 0.8 x design LL plus 0.6 x reverse temperature difference exceeds f_{ctm}; cracked zones reopen in a load test beyond the decompression moment (I_{eff} with $I_{cr}/I_g = 0.35$)	3-35 % of P_e. Uncertainty of the design long-term loss alone is ±30 % of 12.7 %, i.e. about 4 % of P_e; the upper end covers corroded tendons reported in practice
Concrete deterioration (CD)	poorly compacted or chemically attacked concrete in one girder	modulus reduction d_E over a 6-15 m zone or the whole girder, strength reduction 0.9 d_E, voids for UPV, faster carbonation	d_E = 5-30 %; above 30 % the girder would show obvious distress
Flexural cracking (CR)	an overloaded vehicle beyond the design load	EI loss d_s over 4-10 m near mid-span in the load test; in ambient vibration only 20-50 % of it, because cracks in a prestressed girder close again; residual crack 0.05-0.35 mm	d_s = 15-50 % local, i.e. a few per cent of the girder's overall stiffness
Corrosion (CO)	water from a leaking joint or a short drain spout on low-cover concrete of an outer girder; carbonation reaches the links, then strands in poorly grouted ducts	strand area loss c over 2-8 m at a joint end or under a spout, local prestress loss = c, EI loss 0.25c from spalling; wetting, low cover, HCP shift and lower resistivity in the NDT; rust and spalling visible only once enough rust has formed	c = 2-25 %; 80 % of cases on G1/G6, which carry the drains
Bearing deterioration (BR)	hardening of the elastomer with age and debris from a leaking joint packing the bearing	shear stiffness multiplier 5-500 at one or both ends. In the 2-D grillage a horizontal restraint at bearing level becomes a coupled end-rotation spring $k_s e^2$ [1 -1; -1 1], with k_s the series stiffness of substructure, bearings and girder	a bearing locked by debris behaves as several hundred times stiffer in shear; 60 % of cases come with a visibly leaking joint
Local stiffness loss (LS)	a tall vehicle on the service road striking a fascia girder; internal void or delamination	EI loss 15-60 % over 1-3 m anywhere between 5 and 28 m; 75 % visible on outer girders, 40 % on inner girders (otherwise found only by UPV or hammer)	local damage that removes a large part of one section but little of the span stiffness

Healthy campaigns also vary: the age of the bridge (5-30 years), carbonation depth growing with the square root of age, temperature, deck moisture and traffic. A campaign can contain one mechanism (60 %), two at once (15 %) or none (25 %).

10.1 What each mechanism does to the measured quantities

Before generating any random data I swept each mechanism from zero to the top of its range on girder G1 (or on all girders where that is the natural case) with the noise switched off. Figure 14 is the most important figure of the monitoring part, because it shows what the measurements can and cannot see before any machine learning is involved.

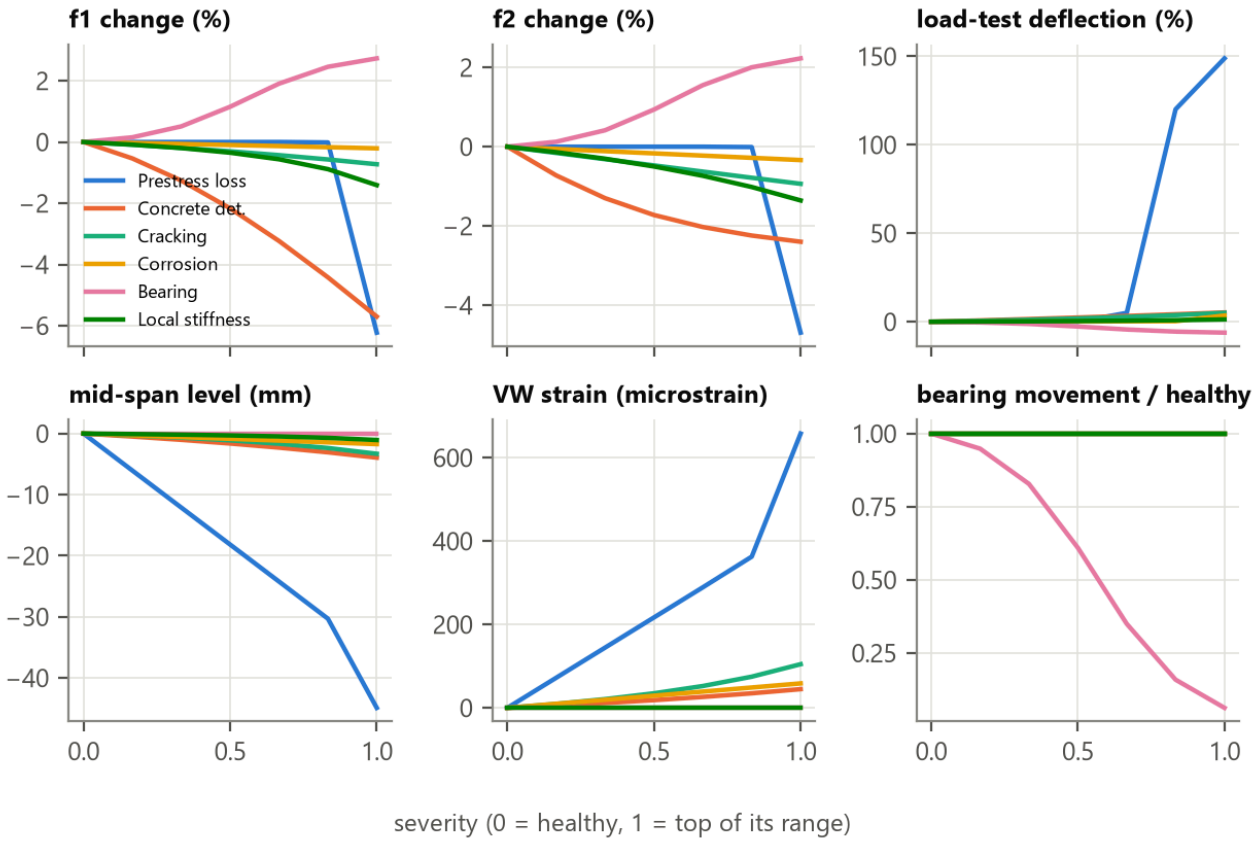


Figure 14. Noise-free response of the twin to each mechanism, from healthy (0) to the top of its range (1).

- **Prestress loss** leaves the frequencies and the load-test deflection unchanged until the girders crack, at roughly 25 % loss here; then the load-test deflection jumps (+149 % at 35 %). Levelling (-18.2 mm at 18 % loss) and VW strain (217 microstrain) respond from the start. This agrees with the theory of bonded prestressed beams, where the axial force does not soften the member (Hamed and Frostig, 2006).
- **Bearing restraint** *raises* the frequencies, by up to 2.7 % for f_1 when both ends lock, because a horizontal force at the bearing level, below the neutral axis, resists the end rotation. A rise in frequency is therefore not proof of a healthy bridge. Restraint at only one end hardly changes the frequency (the free end takes the movement) but shows clearly in the bearing transducers.
- **Concrete deterioration** of a whole girder lowers f_1 by up to 5.7 % and raises the load-test deflection by 5 %; a mechanism that changes stiffness over a large volume is the one global vibration sees best.
- **Cracking, corrosion and local stiffness loss** change the global quantities by a fraction of a per cent to a few per cent. They are local, and the six-girder deck redistributes load around them.

These changes have to be read against the temperature effect: a 10 °C warmer day lowers f_1 by about 1.5 %, which is as much as a 20 % modulus loss of one girder. The brief warns that a change of frequency should not be read as a particular type of damage, and Figure 14 shows why: four mechanisms lower the frequency, one raises it, one leaves it alone, and the environment does both.

11 Synthetic data

The synthetic data are produced by the updated twin (chapter 5), not by the 'true' model of the bridge, exactly as they would have to be in practice: one can only simulate the bridge one has a model of. Every campaign goes through the same chain as a real measurement campaign.

Table 14. Generation of one synthetic campaign

Step	What is generated	Variability and noise
Scenario	0, 1 or 2 mechanisms with random size and location (chapter 10)	25 % healthy, 60 % single, 15 % pairs
Environment	effective bridge temperature from a seasonal-daily model of the site; monsoon deck moisture	16-41 °C; mass +0.6 % in the monsoon, ±0.2 % random
Vibration	modal superposition of 8 modes with moving lorries (bounce 1.3-2.6 Hz, 2-6 %), cars and broadband ambient forces; 12 channels, 30 min, 100 Hz	traffic 3-60 vehicles per 10 min at night, 150-420 by day; damping 1.5 ± 0.2 %; sensor noise 2 µg/√Hz
Identification	the full chain of chapter 9 on each record	f, MAC, damping, a_{RMS}, a_{max} as identified, not as simulated
Load test	4 x 300 kN lorries, deflection and strain of 6 girders	lorry weight ±1 %, 0.05 mm, 3 microstrain
Geometry and VW strain	change beyond the healthy creep prediction	creep-model error 2.5 mm and 25 microstrain (common to all girders), instrument 0.8 mm and 3 microstrain
Bearings	movement per °C at both ends	±0.012 mm/°C
NDT	30 test areas of span S2: rebound, UPV, cover, HCP, resistivity, carbonation, chloride	as-built scatter of strength, cover and carbonation rate; reading errors of each test
Visual inspection	observations with sizes and condition states	hairline cracks missed with probability $e^{-w/0.06}$; condition state ±1 for 15 % of entries; early defects often not visible

3600 campaigns were generated (902 healthy, about 362 for each single mechanism and 521 with two mechanisms), each with 52 raw quantities and its true labels, in `out/data/library.csv`. The examples below are in the format of the brief.

Table 15. Synthetic displacement, strain and frequency for increasing prestress loss (noise-free, 25 °C)

Condition	Load-test deflection G1 (mm)	Load-test deflection G3 (mm)	Strain G1 (microstrain)	Level change (mm)	VW change (microstrain)	f_1 (Hz)
Healthy	4.38	6.31	60	+0.0	+0	3.741
Prestress loss 20 % (all girders)	4.38	6.31	60	-20.8	+248	3.741
Prestress loss 27 % (all girders)	9.29	11.44	183	-28.0	+335	3.741
Prestress loss 33 % (all girders)	12.05	14.18	194	-38.1	+529	3.634

The displacement table makes the point of chapter 10 in numbers: up to 20 % loss the load test and the frequency show nothing, whereas the level and the VW strain have already moved by about 21 mm and 250 microstrain. Once the girders crack (27 %) the load-test deflection increases sharply.

Table 16. Synthetic NDT data along girder G1 of span S2 at year 18

Location	Rebound	UPV (km/s)	Cover (mm)	Carbonation (mm)	HCP (mV)	Resistivity (kΩcm)	Cluster (chapter 13)
S2-G1-Z1	45	4.50	51	16	-241	18	wet / corrosion risk
S2-G1-Z2	43	4.52	53	12	-135	42	sound / dry
S2-G1-Z3	40	4.57	61	11	-149	37	sound / dry
S2-G1-Z4	40	4.41	31	14	-456	12	wet / corrosion risk
S2-G1-Z5	42	4.38	40	12	-436	11	wet / corrosion risk

Along G1 the readings change towards the joint end (Z5): lower resistivity and more negative potentials where the concrete is wet, cover lower than specified and carbonation close to or beyond it. UPV and rebound are less affected, because the concrete itself is sound; corrosion is a problem of the steel and its environment, not of concrete quality.

Table 17. Synthetic vibration data: two deterioration cases and a temperature change

Mode	Healthy (Hz)	Concrete deterioration, G1 -20 % E (Hz)	Both bearing lines restrained x100 (Hz)	Temperature +10 °C (Hz)
f_1 bending	3.741	3.620 (-3.2 %)	3.823 (+2.2 %)	3.685 (-1.5 %)
f_2 torsion	3.932	3.852 (-2.0 %)	4.002 (+1.8 %)	3.873 (-1.5 %)
f_3 transverse	8.815	8.770 (-0.5 %)	8.852 (+0.4 %)	8.684 (-1.5 %)

The brief's example changes f_1 from 3.20 to 3.05 Hz (-4.7 %). Since f varies with √EI, a change of that size on B33 would need a loss of about 9 % of the bending stiffness of the whole deck, more than any single mechanism in my ranges produces, or simply a temperature rise of about 30 °C. The frequency alone cannot tell these apart. The pattern across the modes carries more information than any one of them: deterioration of G1 lowers f_1 more than f_2 and f_3, bearing restraint raises f_1 and f_2 but hardly f_3, and temperature moves all three by nearly the same percentage. That is why all four tracked modes are kept as features.

12 Feature extraction

A feature should have a physical meaning, respond to at least one mechanism, and be stable on the healthy bridge. Every feature here is expressed as a change from the structure's *own* healthy baseline: the frequency residual after the temperature/traffic model of chapter 5, the load-test deflection relative to the commissioning test (corrected to 25 °C with the modulus-temperature relation), levels and VW strain relative to the creep prediction, bearing movement relative to αL/2. This matters because the models are trained on twin data and applied to the real bridge; a 0.5 % error in the twin's absolute frequency would otherwise look like damage.

Table 18. The 33 features used by the AI/ML models

Group	Features	Physical meaning
VIB	f1 change (%), f2 change (%), f3 change (%), f4 change (%), MAC mode 1, MAC mode 2, mode-1 edge asymmetry, damping mode 1	global stiffness, mass, boundary conditions; MAC and edge asymmetry indicate which side changed
LT	load-test deflection change (%), deflection asymmetry G1-G6 (%), max girder deflection change (%), soffit strain change (%), max girder strain change (%)	static stiffness of each girder under a known load; asymmetry locates a soft girder
GEO	mid-span level change (mm), largest sag of a girder (mm), level asymmetry G1-G6 (mm), VW strain change (ue), max VW change (ue)	permanent deformation: loss of precompression (camber, soffit strain) and creep of a softened girder
BRG	min bearing movement ratio, bearing movement A-B (mm/C)	freedom of the bearings; restraint at one end doubles the movement at the other
NDT	min UPV (km/s), min rebound no., most negative HCP (mV), min resistivity (kOhm cm), min cover (mm), carbonation - cover (mm)	condition of concrete (UPV, rebound) and of steel (HCP, resistivity, cover vs carbonation)
VI	flexural crack width (mm), crack along bars/ducts (mm), spalled area (m2), rust-stain locations, girder condition state, bearing condition state, joint leaking	what an inspector sees: cracks, spalls, rust, bearing and joint distress

The brief's suggested vector x = [f_1, f_2, a_{RMS}, δ_{max}, ε_{max}, UPV, rebound] was the starting point. Two of its entries were changed after looking at the data. a_{RMS} and a_{max} depend on traffic far more than on condition (the night-day ratio is about 2), so they are used to *normalise* the frequencies, not as condition features. And δ_{max} was replaced by the change of mean deflection and the G1-G6 asymmetry, which separate a global from a local change. Figure 15 shows, before any classifier, how strongly each feature reacts to each mechanism, as the shift of its mean in units of the healthy standard deviation.

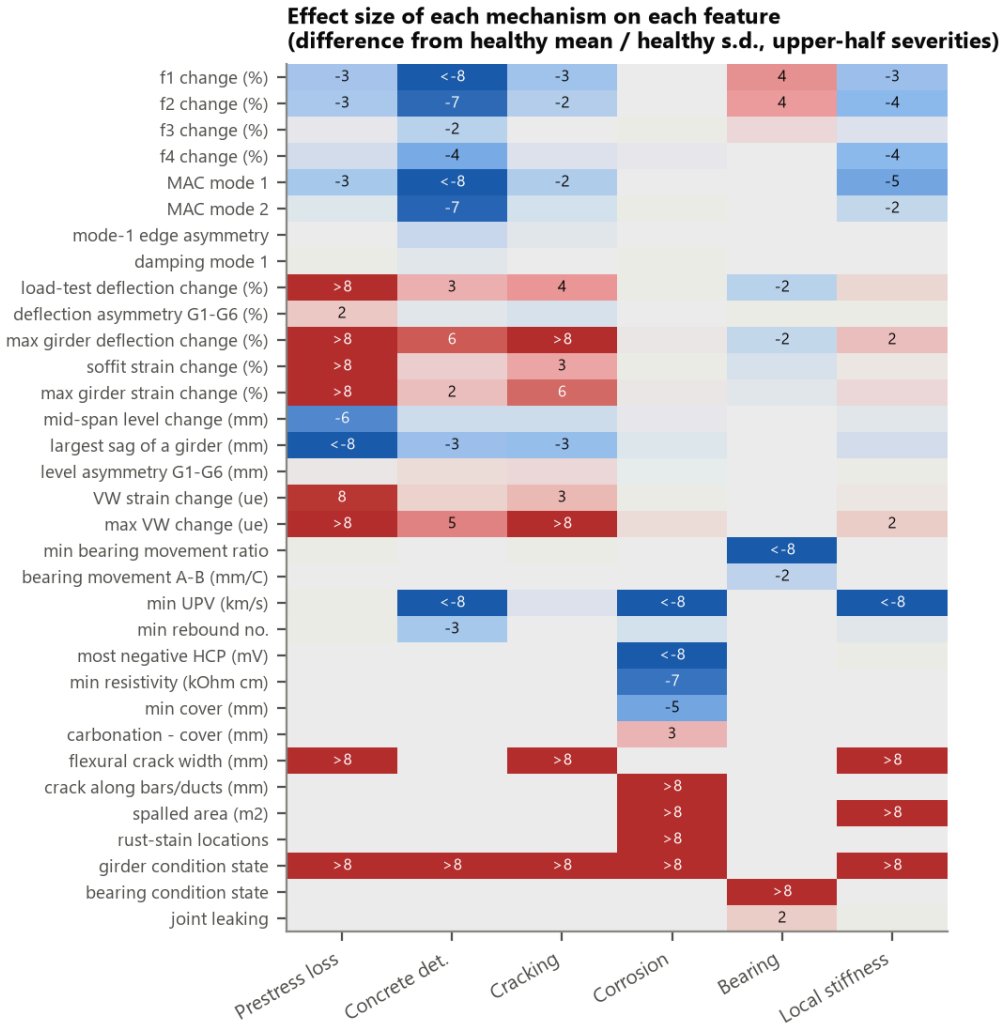


Figure 15. Effect size of each mechanism on each feature (library, upper-half severities). Blue: decrease, red: increase.

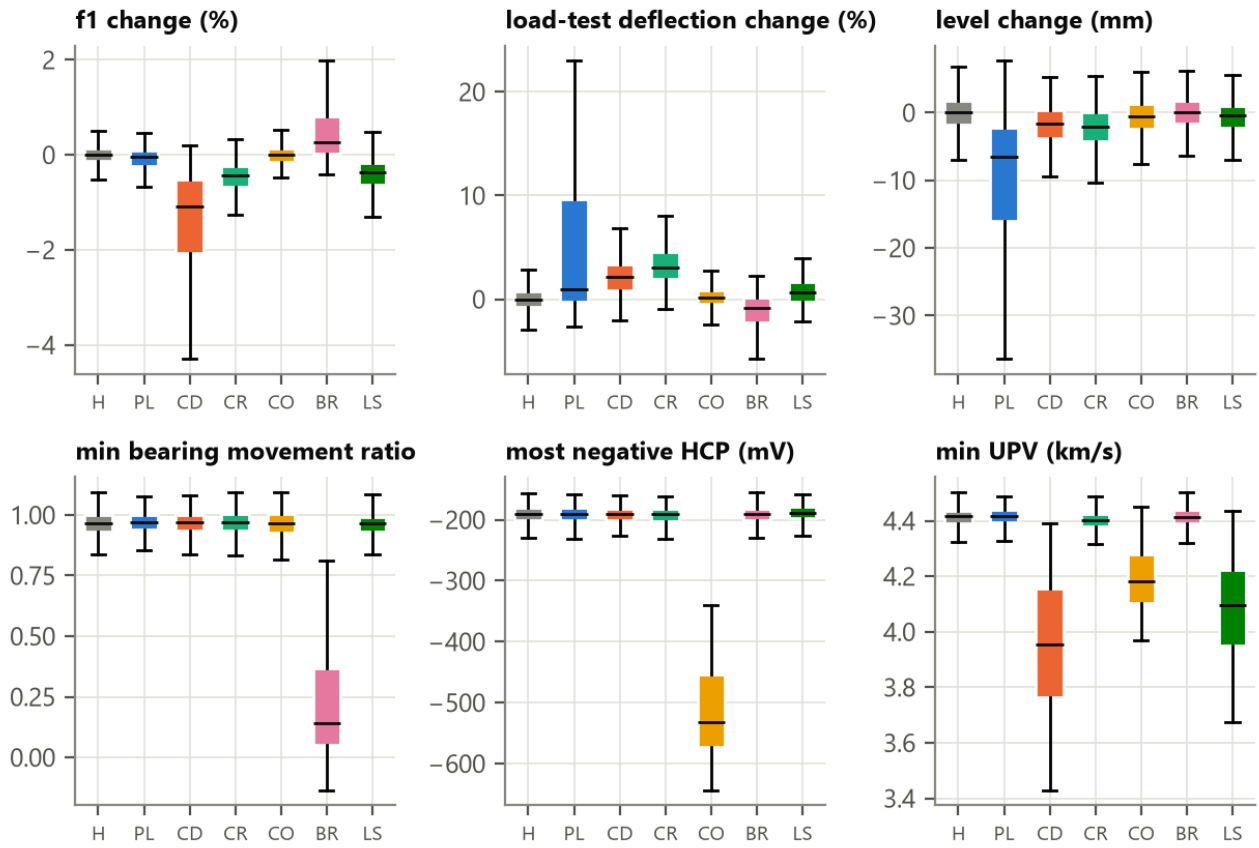


Figure 16. Distributions of six features for healthy (H) and single-mechanism campaigns (boxes: quartiles, whiskers 1.5 IQR).

Reading Figure 15 column by column already gives most of the diagnosis rules an engineer would write: prestress loss moves levels and VW strain; bearing restraint moves the bearing movement ratio, raises f_1 and often comes with a leaking joint; corrosion moves potentials, resistivity and cover against carbonation; concrete deterioration lowers UPV and rebound and, over a whole girder, the frequencies; cracking and local damage show in the load test and in inspection. The value of the ML is to combine these weak and overlapping signals consistently, to put a number on the confidence and to estimate the size.

13 AI/ML-assisted condition assessment

All models were written in numpy and scipy for this work (scikit-learn could not be installed on the computer used, and writing them made the assumptions explicit): a Mahalanobis-distance novelty detector with a shrunk covariance, CART trees and random forests on histogram-binned features, multinomial logistic regression, Gaussian-process regression with one length scale per input, k-means and a Gaussian mixture fitted by EM. The library was split 70/30, stratified by class (2509 training and 1091 test campaigns); every number below is on the test set. The question each model answers is stated with it.

13.1 Is the bridge different from its healthy self? (novelty detection)

The detector is trained on healthy campaigns only, which is the realistic situation: a real bridge has a healthy baseline but no labelled examples of its own damage. The threshold is the 99th percentile of the healthy training distances, i.e. a designed false-alarm rate of 1 %.

Table 19. Share of test campaigns flagged by the novelty detector

Data source	False alarm	Prestress loss	Concrete deterioration	Flexural cracking	Corrosion	Bearing deterioration	Local stiffness loss
Vibration only	1 %	12 %	87 %	45 %	2 %	40 %	60 %
Static + geometry + bearings	1 %	85 %	73 %	89 %	14 %	100 %	29 %
Inspection + NDT	2 %	33 %	92 %	95 %	100 %	85 %	98 %
Continuous monitoring (no closure)	2 %	79 %	89 %	76 %	12 %	100 %	60 %
All sources fused	3 %	86 %	96 %	100 %	100 %	100 %	98 %

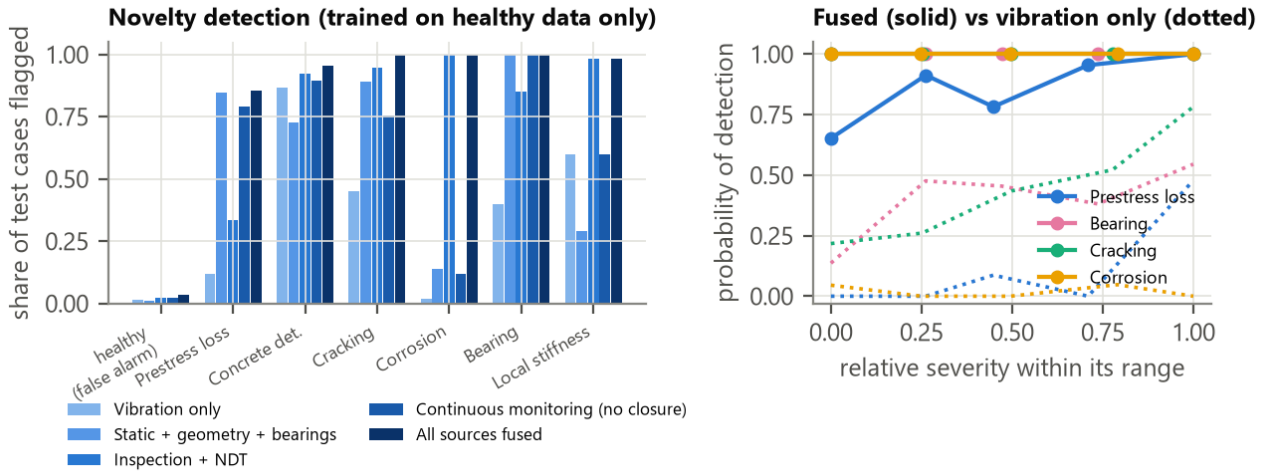


Figure 17. Novelty detection by data source, and probability of detection against severity (fused: solid; vibration only: dotted).

With vibration alone the detector finds most cases of concrete deterioration (87 %) and local stiffness loss (60 %), fewer bearing cases (40 %, only those restrained at both ends), and almost no prestress loss (12 %) or corrosion (2 %). This is Figure 14 again, now with noise and environment: vibration monitoring alone is the wrong tool for the two mechanisms that most often govern the life of a post-tensioned bridge. Adding levelling, VW strain and bearing movement, all of which can run without closing the bridge, lifts the detection of prestress loss to 79 % and of bearing problems to 100 %. With all sources fused, detection is 86 % or better for every mechanism at a false-alarm rate of 3 %; the misses are the smallest severities (Figure 17, right).

13.2 Which mechanism? (supervised detection and data fusion)

For each mechanism a random forest (100 trees, depth 12, √p features per split) was trained to answer 'is this mechanism present?', so two mechanisms can be reported at once. The same model was trained on each data source, which is the data-fusion study: what does each source add?

Table 20. ROC-AUC / F1 per mechanism and data source (test set)

Data source	Prestress loss	Concrete deterioration	Flexural cracking	Corrosion	Bearing deterioration	Local stiffness loss	macro-F1
Vibration only	0.57 / 0.04	0.92 / 0.66	0.85 / 0.31	0.57 / 0.00	0.83 / 0.54	0.81 / 0.39	**0.32**
Static + geometry + bearings	0.91 / 0.78	0.82 / 0.46	0.92 / 0.64	0.64 / 0.00	1.00 / 1.00	0.67 / 0.07	**0.49**
Inspection + NDT	0.73 / 0.29	0.97 / 0.84	0.97 / 0.87	1.00 / 1.00	0.97 / 0.93	0.99 / 0.86	**0.80**
Continuous monitoring (no closure)	0.88 / 0.74	0.92 / 0.68	0.88 / 0.47	0.64 / 0.00	1.00 / 1.00	0.83 / 0.40	**0.55**
All sources fused	0.93 / 0.79	0.98 / 0.84	1.00 / 0.94	1.00 / 1.00	1.00 / 1.00	0.99 / 0.81	**0.90**

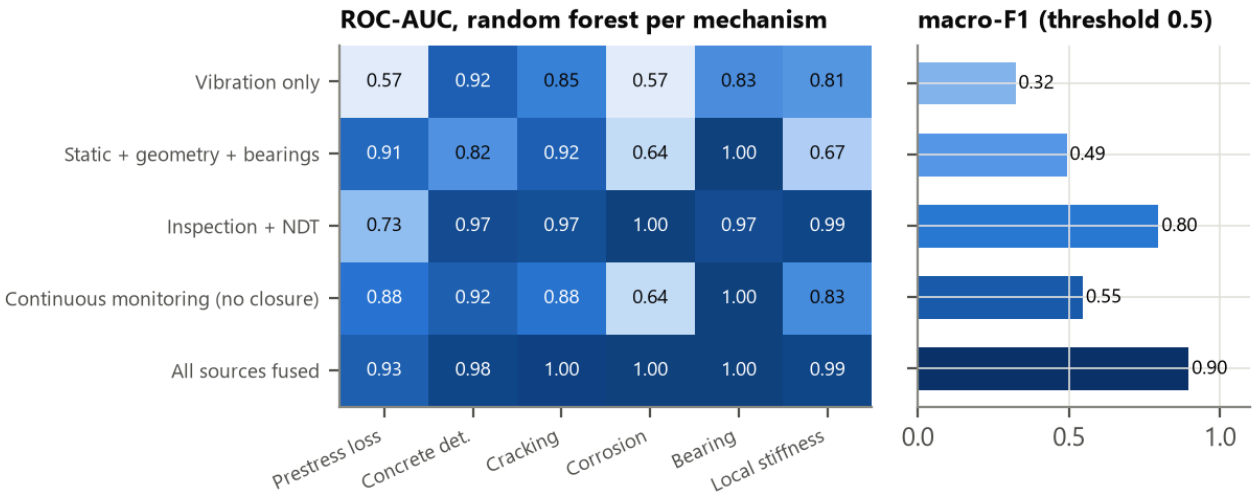


Figure 18. ROC-AUC per mechanism and data source, and the macro-averaged F1 score.

The macro-F1 rises from 0.32 with vibration alone to 0.49 with static, geometric and bearing data, 0.80 with inspection and NDT, and 0.90 with everything. No single source is enough, and the sources complement each other in a way that follows the physics: NDT and inspection see corrosion and surface damage, levelling and VW strain see prestress, the bearing transducers see the bearings, the load test sees cracking. A logistic regression on the fused features reaches AUC 0.88-1.00, close to the forest; the choice of features mattered far more than the choice of algorithm.

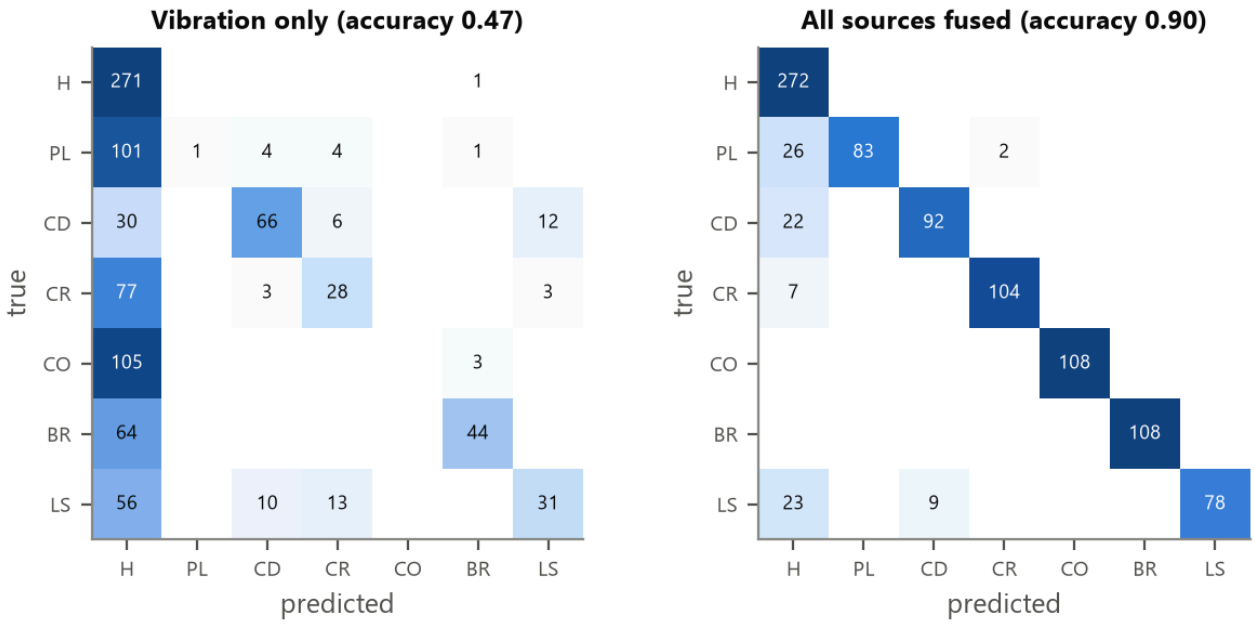


Figure 19. Dominant mechanism on healthy and single-mechanism test campaigns: vibration only against all sources fused.

The confusion matrices show what goes wrong. With vibration alone, prestress loss and corrosion are mostly called healthy, and cracking, local stiffness loss and concrete deterioration are confused with each other, because all three lower the frequencies a little and differ mainly in where and how much. With all sources, the remaining errors are early prestress loss and the smallest cases of cracking and concrete deterioration, i.e. cases that are hard to see by any means.

13.3 How much prestress has been lost? (regression with uncertainty)

The size of a prestress loss decides between monitoring and strengthening, so a point estimate without its uncertainty is of little use. A Gaussian-process regression with inputs worst-girder sag, mean level, largest and mean VW strain, load-test deflection change and flexural crack width was trained on campaigns with prestress loss. On the test set it gives a mean absolute error of **2.6 %** of P_e (R^2 = 0.83), and the 95 % band contains the true value in 92 % of cases, so the stated uncertainty is honest (Figure 20). A random forest gives a similar error (2.7 %). A purely physical inverse from the twin's sensitivities (1.04 mm and 12.4 microstrain per 1 % loss) on the worst instrumented girder is worse (MAE 6.4 %), mainly because it cannot see losses in G2 and G5, which have no gauges, and is disturbed by other mechanisms in the two-mechanism cases. The learned length scales tell which inputs the GP relies on: the shortest belong to load-test deflection change (%) and max VW change (ue).

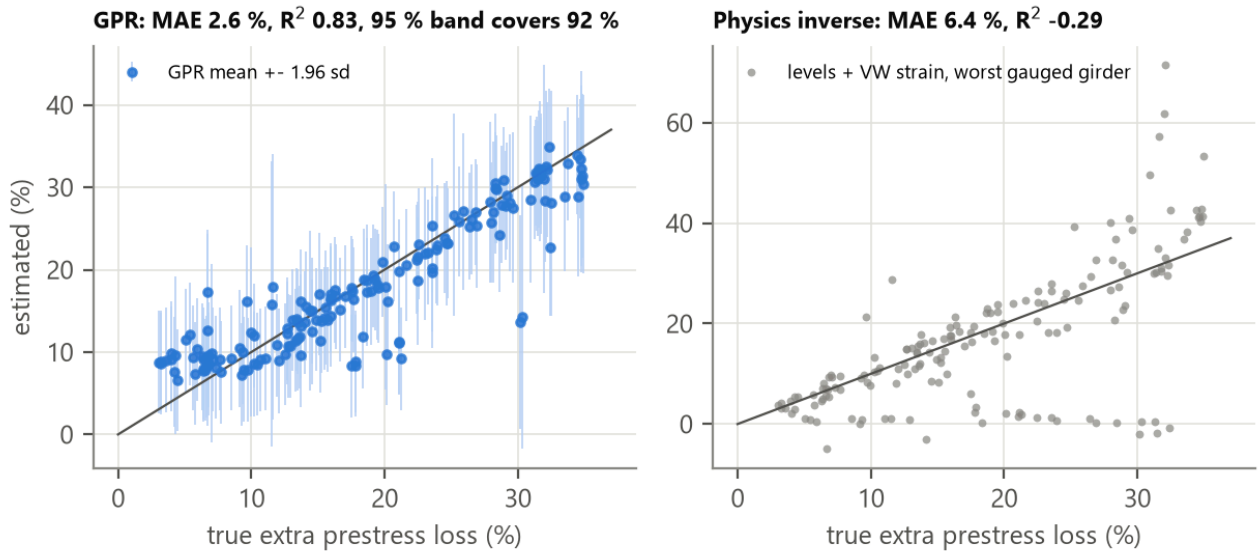


Figure 20. Estimated against true extra prestress loss on test campaigns: Gaussian process and physics-only inverse.

13.4 Where? (localisation)

A random forest on the girder-wise load-test deflections and strains, levels, VW strains and mode-shape asymmetry identifies the affected girder in **70 %** of single-mechanism test cases (prestress loss 75 %, concrete deterioration 78 %, flexural cracking 87 %, corrosion 45 %, local stiffness loss 64 %). Corrosion is located worst because its global signature is small and often near a support, where the load test is insensitive; for corrosion the location comes from the NDT grid instead (section 14.3).

13.5 Do the models use the physics? (importance)

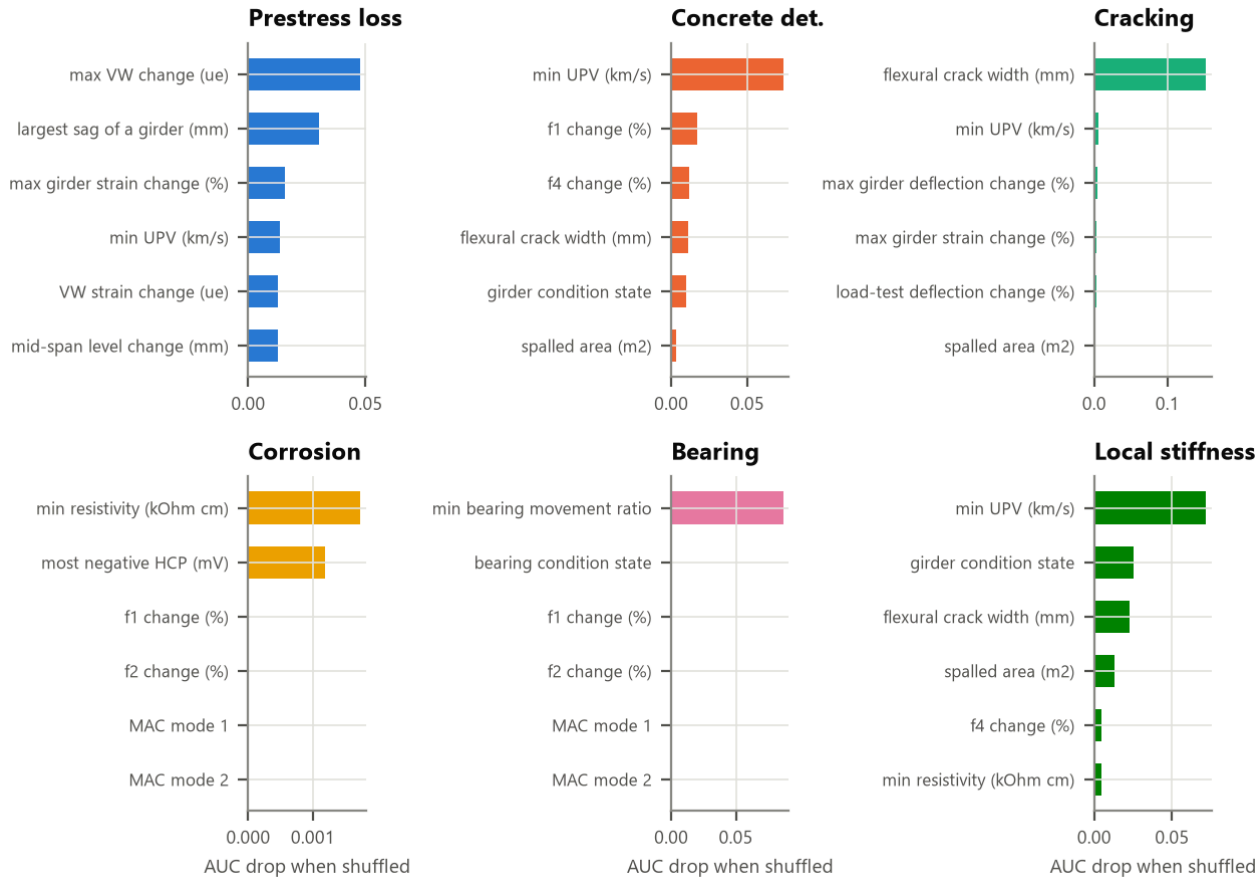


Figure 21. Permutation importance (drop of test AUC when a feature is shuffled) for the fused random forests.

The permutation importances (Figure 21) agree with the effect sizes of Figure 15 and with the mechanisms of chapter 10: the prestress model relies on levels and VW strain, the bearing model on the bearing movement ratio, the corrosion model on potentials, resistivity and rust, the

concrete model on UPV and rebound. I regard this agreement as a necessary condition for trusting the models; a model that detected corrosion from the damping of mode 1, say, would have learned a quirk of the simulation.

13.6 Limitations

- All data are synthetic. The scores measure how well the method works *if the twin and the noise models are right*; on a real bridge they would be lower and would need recalibration with the bridge's own data.
- The twin is a linear grillage. Bond loss, crack interface behaviour, bearing friction and nonlinear soil-structure effects are represented only through simple rules.
- Labels are perfect in the library. In practice the label of a past campaign is often unknown, which is why novelty detection on the healthy baseline, not the supervised models, is the first line of the assessment.
- Six mechanisms were considered. A different mechanism (fatigue of a strand, scour of a pier) would appear as 'novel but unclassified', which is the right output for an unknown situation.

14 Year-18 assessment of B33

This chapter puts everything together for the bridge itself. The 'real' bridge (the hidden model of chapter 5) was aged for 18 years with a deterioration history that was not used by the assessment. Monthly monitoring records (night-time vibration, bearing movement, VW strain) and annual levels were produced for the whole period, and in year 18 a principal inspection with NDT of all spans and a load test was carried out. The assessment below uses only these measurements, the baseline of chapter 5 and the models of chapter 13.

14.1 What the monitoring showed

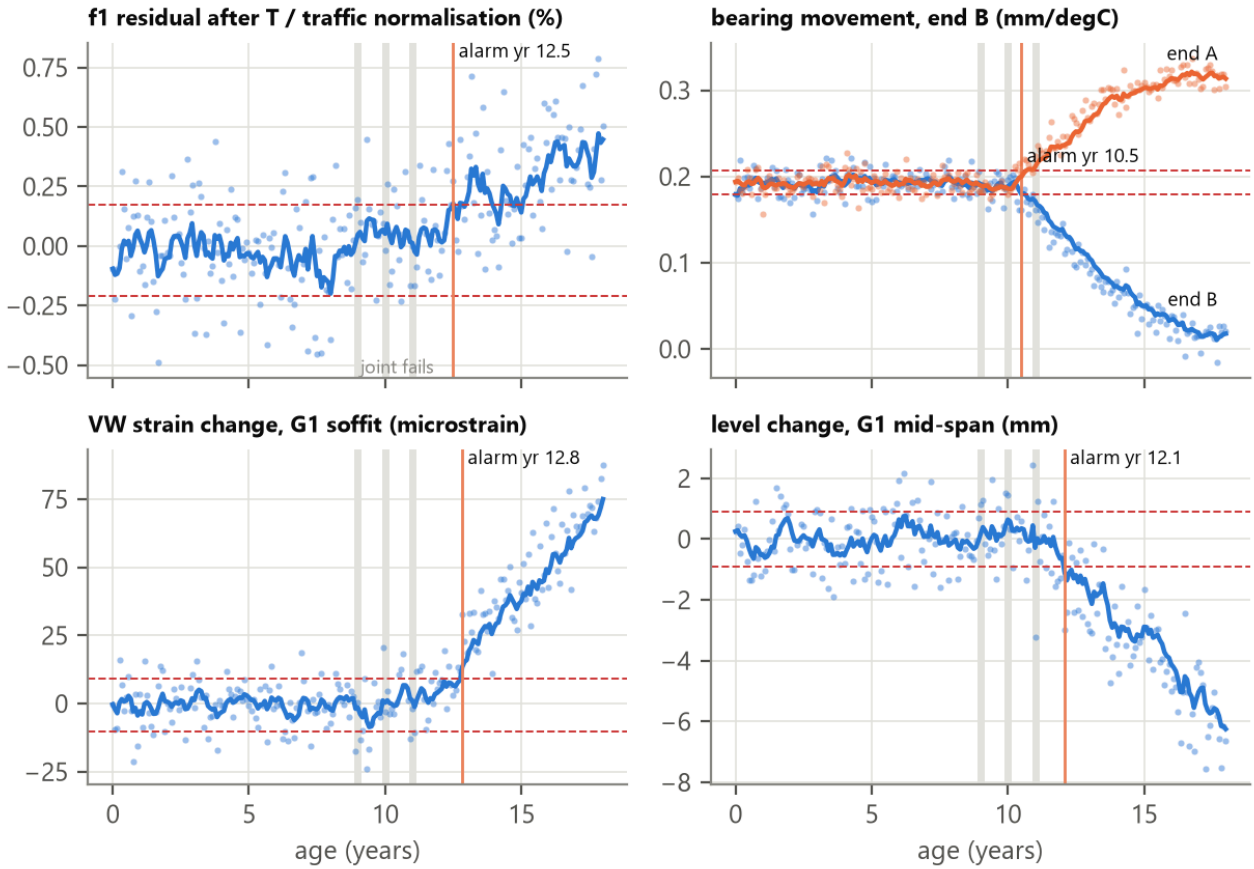


Figure 22. 18 years of monitoring of span S2: monthly values and EWMA (λ = 0.2) with ±3σ limits from the first two years. Grey bars: onset of the hidden changes.

Table 21. Trend identification on the monitoring record

Monitored quantity	First EWMA alarm (year)	Theil-Sen slope, last 3 years (per year)
f1 residual	12.5	+0.030
f2 residual	13.2	+0.039
bearing B movement	10.5	-0.009
bearing A movement	10.6	+0.004
VW strain G1	12.8	+12.575
level G1	12.1	-1.143

The bearing transducers gave the first and clearest signal (year 10.5): the movement at end B fell towards zero and that at end A rose, the signature of a bearing that is being locked at one end while the other end takes over the movement. The f_1 residual also alarmed, in year 12.5, but *upwards*: the span had become stiffer. On its own that could pass for a healthy sign or a temperature-model error; next to the bearing record it is the restraint of chapter 10. The G1 level dropped (alarm in year 12.1) and the VW gauge in G1 drifted towards tension (year 12.8), the pattern of lost precompression in one girder. All alarms came one to two years after the hidden onsets, which is about what EWMA charts with these limits can do with monthly data. With vibration alone the record would have said 'stiffer', which is the wrong way round for a deteriorating bridge.

14.2 Principal inspection and NDT, year 18

Table 22. Main observations of the year-18 inspection (condition state 3 and worse)

Span	Component	Location	Observation	Size	CS	Note
S2	G1	Z4	Cracking	0.39 mm	5	longitudinal, along bars / duct, rust in crack
S2	G1	Z5	Cracking	0.45 mm	5	longitudinal, along bars / duct, rust in crack
S2	BRG	G3-B	Bearing distress		5	hardened / restrained, debris around bearing
S2	BRG	G2-B	Bearing distress		4	hardened / restrained, debris around bearing
S2	BRG	G4-B	Bearing distress		4	hardened / restrained, debris around bearing
S2	BRG	G5-B	Bearing distress		4	hardened / restrained, debris around bearing
S2	BRG	G6-B	Bearing distress		4	hardened / restrained, debris around bearing
S2	EJ	EJ-P2	Leakage		4	water and silt onto pier cap
S2	G1	Z4	Spalling	0.18 m2	3	
S2	G1	Z4	Leakage	damp patches / efflorescence	3	
S2	G1	Z5	Corrosion staining		3	
S2	G1	Z5	Leakage	damp patches / efflorescence	3	
S3	G1	Z1	Leakage	damp patches / efflorescence	3	
S2	EJ	EJ-P1	Joint distress		3	seal torn, debris packed in gap

The bridge condition index is **38** out of 100. The worst component groups are G (CS5), BRG (CS5), EJ (CS4), DR (CS3). All the serious findings are within a few metres of the expansion joint over pier P2: the torn seal, silt around the S2 bearings at end B and the S3 bearings at end A, leakage and spalling on the pier cap, a short drain spout and rust-stained splitting cracks and spalling at the anchorage end of G1. The inspector rated two of these cracks, 0.39 and 0.45 mm along the duct line in zones Z4-Z5, as CS5; by the width rule they would be CS4. I kept the inspector's rating in the record, because rust coming out of a crack along a tendon duct next to an anchorage is exactly the warning sign that preceded the Ynys-y-Gwas collapse (Woodward and Williams, 1988), but the category of the bridge was decided after the structural check of section 14.3, not from that single rating.

The 90 NDT test areas were clustered with a Gaussian mixture on standardised UPV, rebound, half-cell potential, log-resistivity and cover minus carbonation depth. The BIC was lowest for 2 components (Table below), so two groups were kept: 83 areas of sound, dry concrete and 7 areas of wet concrete with corrosion risk. All 7 lie at girder ends next to the joints over the two piers (S1-G1-Z5, S1-G6-Z5, S2-G1-Z1, S2-G1-Z4, S2-G1-Z5, S2-G6-Z1, S3-G1-Z1). Inside that group the ASTM C876 bands do the rest of the work: only S2-G1-Z4, S2-G1-Z5 have potentials below -350 mV together with resistivity near 10 kΩcm, i.e. probable active corrosion; the others are wet but still passive. k-means with the same number of groups agrees with the mixture for 98 % of location pairs.

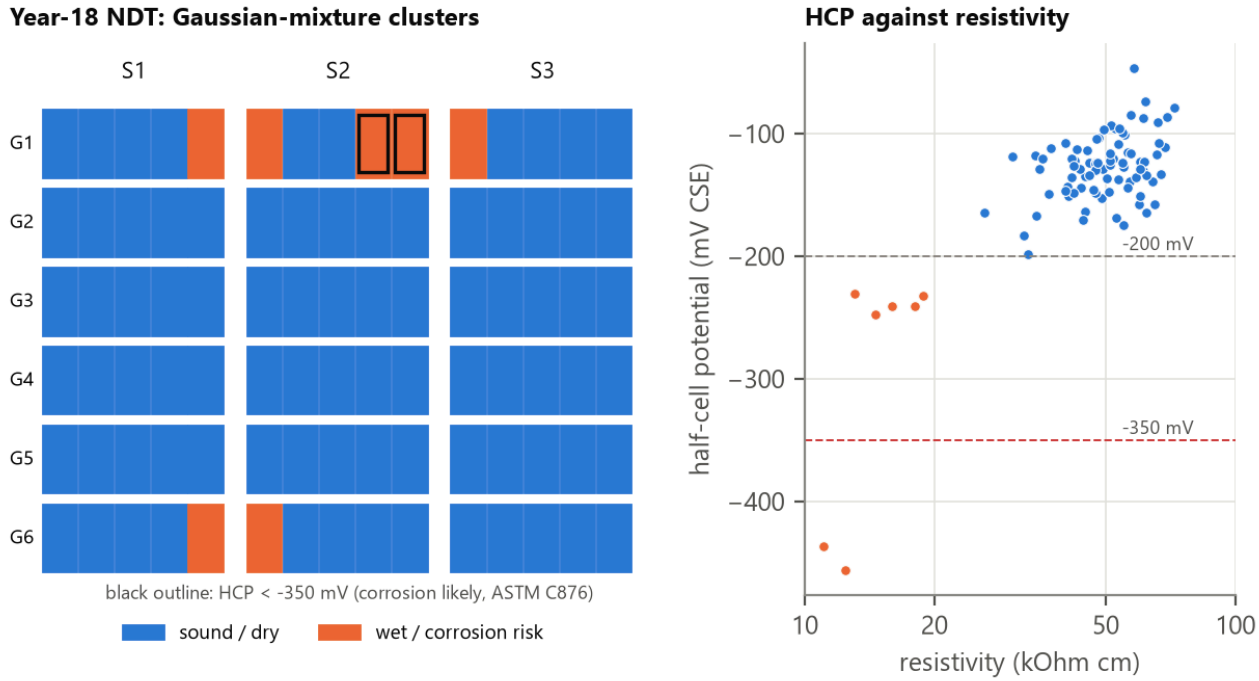


Figure 23. Year-18 NDT survey of all girders, grouped by a Gaussian mixture; right: potential against resistivity with the ASTM C876 bands.

Table 23. NDT clusters (mean values)

Cluster	Areas	UPV	Rebound	HCP (mV)	Resistivity	Cover - carbonation (mm)	Members
sound / dry	83	4.56	43.9	-127	51	+39	83 areas
wet / corrosion risk	7	4.57	41.6	-298	15	+34	S1-G1-Z5, S1-G6-Z5, S2-G1-Z1, S2-G1-Z4, S2-G1-Z5, S2-G6-Z1, S3-G1-Z1

The clustering is useful because it is unsupervised: it does not know about the joints, yet it separates the wet end zones next to both piers from the rest of the bridge, which tells the engineer that the second joint (P1) also needs attention. The engineering bands then single out the G1 end zone

of S2. The concrete itself (UPV, rebound) is sound almost everywhere, so this is a problem of water and steel, not of concrete quality.

14.3 AI/ML output and the engineer's reading

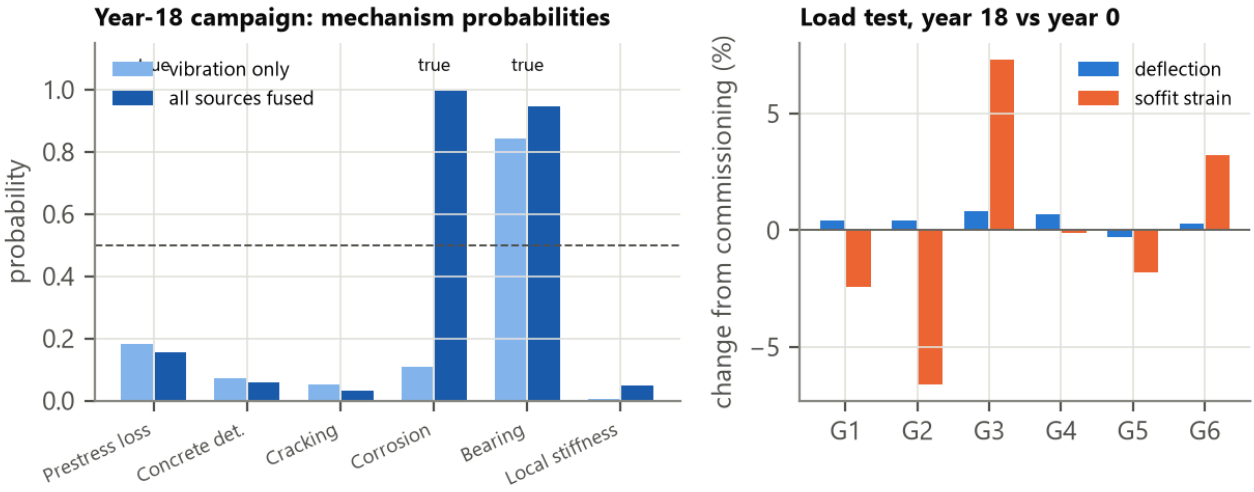


Figure 24. Year-18 campaign: mechanism probabilities from the fused and the vibration-only models; change of the load-test response per girder since commissioning.

The fused Mahalanobis distance is 186 against a threshold of 6.9: the bridge is clearly no longer its healthy self (most of the distance comes from inspection features that are always zero on the healthy bridge, such as rust stains). The fused models give the highest probabilities to corrosion (1.00), bearing deterioration (0.95), prestress loss (0.15). The vibration-only models give bearing deterioration 0.84, prestress loss 0.18: they find the bearings, which stiffen the span, and nothing else. The localisation model gives G1 a probability of 0.55.

The prestress estimate needed more care. My first version of the Gaussian process, which took the largest crack width of any kind as an input, gave **18 %** for this campaign. It had been misled by the 0.45 mm rust-stained splitting cracks at the joint end: in its training data a visible crack on a girder almost always meant a flexural crack, which appears only after a loss of about 25 %. After splitting the crack feature into flexural cracks and cracks along bars or ducts and retraining, the model gives 9.5 ± 6.3 % (95 % band), and its test error also improved. Because the levels and the VW strain moved mainly in G1, I checked this by inverting the G1 measurements directly with the twin sensitivities (1.04 mm and 12.4 microstrain per 1 % loss), which gives 4.4 ± 3.2 % for G1 (levelling 2.3 %, VW strain 5.8 %). The two estimates overlap; I used the direct G1 estimate for the structural check because it rests on two measurements whose physics is clear, and carried the GPR value as an upper indication.

Is this physically reasonable? I checked it against the mechanisms rather than accepting it. A leaking joint explains the bearing distress at both spans next to it, the wet concrete and low resistivity at the girder ends, and, through the anchorages at that end, a possible loss of prestress in G1; the slight rise of f_1 fits restrained bearings; the absence of any change in UPV and rebound rules out a concrete problem; the load test changed little, which is consistent with a girder that has lost some precompression but has not cracked. The ML output is therefore consistent with one cause, the joint, and three consequences. What it cannot tell is whether the strands inside the ducts at the G1 anchorage are corroding; only an intrusive tendon investigation can show that, and the maintenance plan has to include one.

Structural check with the estimated loss. Using the G1 estimate (4.4 %), the soffit margin under the rare combination becomes +0.09 MPa on the design basis (0.9 P_m) and +2.33 MPa with mean prestress, against cracking at $-f_{ctm}$; the quasi-permanent soffit stress is still +4.6 MPa compression. The ULS flexural capacity at mid-span is not affected by a loss of force (the strands yield anyway), $M_{Ed}/M_{Rd} = 0.76$. At the corroded B end the vertical prestress component falls, and the net design shear at 2.75 m rises from 690 to 757 kN, still below $V_{Rd,max} = 1333$ kN and covered by the adopted links. The girder is therefore safe today, but it is close to losing its design margin against cracking, and its condition will worsen if water keeps reaching the anchorage.

14.4 Check against the hidden history

The history used to create the data was: joint seal at pier P2 failing in year 9; bearing restraint at end B (stiffness x100) and slight hardening at end A from year 10; corrosion of G1 at the B end, x = 28.5-33.0 m, strand area loss 8.4 % from year 11; extra prestress loss in G1 of 6.3 %. The assessment found all three mechanisms and their common cause, located them correctly, and estimated the prestress loss of G1 within its stated uncertainty. It did not, and could not, measure the strand corrosion inside the ducts; it pointed to where that has to be investigated.

15 Maintenance scheme

The maintenance decision combines all sources, as the brief asks: visual inspection + NDT + structural response + vibration monitoring + AI/ML. In my scheme the ML output is one line of evidence among them. An action is triggered by the evidence itself, a crack, a potential map, a bearing that does not move, and the ML result mainly sets the priority and tells what to check next. A probability of 0.8 for 'prestress loss' is not a reason to strengthen a girder; it is a reason to measure the force.

15.1 Decision framework

Condition assessment VI + NDT + response + vibration + AI/ML	→	No alarm, CS ≤ 2: **routine monitoring** pre/post-monsoon inspection, monthly data review	or	Alarm or CS ≥ 3: **detailed investigation** special inspection, targeted NDT, assessment	→	**Maintenance / repair** remove the cause first, then the damage	→	**Reinspection and monitoring** new baseline after works

Table 24. Decision matrix used for B33

Evidence pattern	Interpretation	Action
All sources within baseline, CS <= 2	healthy, normal ageing	routine inspection and monitoring
Frequency residual alarm only, nothing else	environment / mass / traffic effect or early stiffness loss	check temperature model and records, repeat in 1 month; do not act on frequency alone
Frequency rise + bearing movement drop	bearing restraint (seizure / debris)	inspect bearings and joint, clean, replace
Level sag + VW tension, no cracks	prestress loss beyond the creep prediction	lift-off / stress release test, assessment
Level sag + load-test softening + flexural cracks	decompression lost, girder cracked	assessment, load management, external PT
HCP < -350 mV + low resistivity + carbonation at cover	active corrosion of links / bars	patch repair, coating, inhibitor
Corrosion signs near anchorages / joint ends	possible strand corrosion in ducts	GPR + impact-echo + borescope, assessment
Low UPV + low rebound over a zone	concrete deterioration / honeycombing	cores, petrography, repair or encasement
Local deflection / strain jump at one girder + spalling	local stiffness loss (impact, delamination)	special inspection, repair

15.2 Maintenance plan for B33 after the year-18 assessment

The assessment places the bridge in the category **Serious - detailed investigation**. The plan (Table below) first removes the cause, water at the pier P2 joint, then repairs the damage it has done, then measures what cannot be inferred, and finally re-baselines the monitoring so that the repaired bridge is again compared with itself.

Table 25. Actions, time frames and the evidence behind each

Action	When	Triggered by	Notes
Clean and extend drainage spouts	within 1 month	blocked / short drainage spouts wetting the outer girder (VI)	spouts discharge clear of girder soffit, add drip groove
Replace strip-seal expansion joint	within 3 months	expansion joint seal leaking (VI)	stops water and silt reaching bearings, anchorages and pier cap
Jack deck and replace / reset bearings	within 6 months	practically no bearing movement at one end + bearing distress (VI)	jacking points on end diaphragm (designed for this), lift <= 10 mm
Concrete repair: remove carbonated / delaminated cover, treat bars, polymer-modified mortar	within 6 months	HCP -456 mV with resistivity 11 kOhm cm, spalling/rust (NDT + VI)	after marking out by hammer tapping and half-cell map
Anti-carbonation coating on repaired and low-cover zones	after repair	low cover zones	extends time to re-initiation
Migrating corrosion inhibitor on wet, low-resistivity zones	with repair	wet zones with potentials in the uncertain band	where HCP is in the uncertain band
Tendon investigation: GPR to locate ducts, impact-echo for voids, borescope through drilled ports at anchorages	within 3 months	corrosion near tendon anchorages / ducts, ML p(CO) = 1.00	only direct way to confirm strand corrosion inside the duct
Prestress check: lift-off or saw-cut (stress-release) test on the suspect girder	within 6 months	estimated extra prestress loss 4.4 +- 3.2 %, p(PL) = 0.15; trend alarms: VW strain G1, level G1	measures the remaining force rather than inferring it
Detailed structural assessment with measured losses (SLS decompression, ULS, shear at the end zone)	within 3 months	rare-combination soffit margin +0.09 MPa with the estimated loss	decides whether strengthening is needed
Epoxy injection of cracks wider than 0.2 mm, sealing of finer cracks	within 6 months	crack width 0.45 mm	after the cause is removed
Additional instrumentation: crack gauges, extra VW gauges at the end zone, bearing LVDTs on all girders	within 1 month	monitoring alarm: f1 residual, f2 residual, bearing B movement, bearing A movement, VW strain G1, level G1	to watch the suspect area until the repair
Special inspection by a bridge engineer (under-bridge unit, close visual of all girders)	within 2 weeks	serious defect or monitoring alarm	
Re-inspection and repeat monitoring campaign after the works	6 months after works	close the loop: confirm the effect of the works	new baseline for the repaired bridge

Sequence and justification. Drains and the joint come first because every other defect here is a consequence of water; repairing concrete or bearings while the joint still leaks would waste the repair. The bearings of S2 end B (and S3 end A, same joint) are replaced by lifting each span on jacks placed under the end diaphragm, which was designed as a jacking beam; the lift is limited to 10 mm and the joint is closed to traffic for one night per span. Concrete repair at the G1 end zone and the pier cap follows the half-cell map, with an anti-carbonation coating on repaired and low-cover areas and a migrating corrosion inhibitor where potentials are in the uncertain band. The tendon investigation (GPR to locate the ducts, impact-echo along them, borescope through a small drilled port at the anchorage and at duct high points) and a lift-off or stress-release test on G1 decide the last step: if the measured force confirms a loss that removes the margin against cracking, external post-tensioning (two straight 12T15 tendons per girder with deviators at the mid-span cross girder and anchors on the end diaphragms) restores it; if not, the girder is kept under closer monitoring.

Inspection and monitoring intervals. Until the works are complete: routine inspection monthly until works, monitoring continuous, weekly review; principal inspection after works, then 1 year. Crack gauges and two extra VW gauges at the G1 end zone and bearing transducers on all girders of S2 are added at once and kept for at least two years after the repair. After the works, the commissioning steps of chapter 5 are repeated (ambient test, load test, NDT of the repaired zones) and become the new baseline; the bridge returns to the routine regime (pre/post-monsoon routine inspections, principal inspection every three years, quarterly review of the monitoring) when two years of data after repair stay inside the new limits.

Longer-term improvement. Each of the four expansion joints is a source of water. When the S2-S3 joint is replaced, it is worth considering a continuity (link) slab over the piers, which would leave only the two abutment joints. The drain spouts of all spans should be extended at least 100 mm below the girder soffit with a drip groove, which costs little and removes the second source of wetting found in this assessment.

16 Use of AI tools and verification

As permitted by §1.15 of the brief, I used an AI assistant (Claude, by Anthropic) throughout: to discuss the problem and the approach, to write and debug most of the Python code, to generate the synthetic data, and to draft and edit parts of this report. The assumptions, the engineering choices and the interpretation are mine, and every result was checked before it was used. The brief says that an AI-generated result is not a verified engineering result; the most useful part of the interaction was the list of things that were wrong at first and were changed after checking:

- The first cable layout swung two cables sideways inside the 300 mm web, which is impossible; the profiles were redrawn so the lateral swing happens inside the bottom bulb, and a clash check along the whole girder was added.
- Stressing all four cables at 10 days gave 18.6 MPa at the soffit against a limit of 17.3 MPa; two-stage stressing was adopted.
- The first vibration model gave peak accelerations of 170 mg because the lorry bounce frequencies overlapped the first mode and the ambient forcing was too strong; both were recalibrated until the night-time RMS (0.6-0.8 mg) and the traffic peaks (10-25 mg) were of the order reported for bridges of this size.
- With 10-minute records and peak picking the scatter of f_1 was ±0.5 %; 30-minute records with a fitted spectrum reduced it to about 0.1 %.
- The AutoMAC check showed that the first sensor layout could not separate two pairs of modes; two accelerometers were added.
- Identifying the commissioning test with the design-model shapes biased f_2 by 3.6 %; the commissioning test is now identified model-free (FDD).
- The first synthetic inspection saw every hardened bearing and every 2 % corrosion case, which made inspection look perfect; detection was made to depend on severity, and leaking joints were also allowed on sound bridges.
- The first prestress-loss model mistook corrosion cracks for flexural cracks and estimated 18 % loss for the year-18 campaign; the crack feature was split by crack type and the result checked against a direct physical inversion (section 14.3).
- The novelty detector failed at first on the inspection features (singular covariance, because a healthy bridge never has rust stains); a small floor on the covariance was added and its effect on the false-alarm rate checked.
- The impact factor for 70R wheeled vehicles on a 33 m span, the IRC:6 footway load for spans over 30 m and the three-lane reduction were checked against the code text rather than taken from the assistant's first answer.

Appendix B lists 20 independent checks of the computations by simpler methods (hand formulas, a separate influence-line calculation, Courbon's method, beam-frequency formulas, recovery of the known parameters); 20 of 20 pass their tolerances. The span, the configuration, the prestressing system, the deterioration scenarios and values, the synthetic data and the maintenance scheme were all developed for this report, as the brief requires for an individual problem (§1.16).

17 Conclusions

- A 33 m span with six 1.80 m post-tensioned M45 girders at 2.70 m, four 12T15 cables each and two-stage stressing satisfies IRC:6 / IRC:112 with the soffit uncracked under the rare combination (margin about 1 MPa) and $M_{Ed}/M_{Rd} = 0.76$. The girder number was set by the limited construction depth rather than by material quantity.
- A healthy baseline must be measured, not calculated. The design model was 7-10 % off in frequency; after updating three parameters on the commissioning tests the twin reproduced them within 0.2 %, and one year of data gave the temperature and traffic normalisation without which frequencies are not usable here (4 % seasonal swing).
- Vibration monitoring alone is poor at detecting the mechanisms that usually decide the life of a post-tensioned bridge, prestress loss and corrosion. Levelling, embedded VW strain and bearing transducers are cheap and do it much better; NDT and inspection are indispensable for corrosion.
- Data fusion, not the algorithm, made the AI/ML useful: macro-F1 0.32 with vibration, 0.90 fused; the models' feature importances agree with the physics, and the Gaussian process gives prestress loss with an honest uncertainty.
- Applied to the bridge at year 18, the assessment traced the defects to one cause (a leaking joint), located them, estimated the prestress loss of the affected girder and produced a maintenance plan that removes the cause, repairs the damage, measures what the monitoring can only infer, and re-baselines the bridge.

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Appendix A. Files and how to reproduce the results

File / folder	Content
psc/inputs.py	every adopted dimension, material, load and criterion
psc/section.py, grillage.py, vehicles.py	section properties, grillage FE (static, influence surfaces, modal), IRC vehicles and worst-position search
psc/prestress.py, design.py	cable geometry, losses, stage stresses, ULS, shear, bearings, girder-number study
psc/twin.py, dynamics.py	digital twin (state → responses), vibration simulation, FDD and modal-filter identification
psc/condition.py, scenarios.py, features.py	inspection and NDT models, deterioration scenarios, campaign generator, feature extraction
psc/mlkit.py, maintenance.py	ML methods (numpy/scipy) and the decision rules
run_01_design.py ... run_05_case_study.py	the five steps of the study, in order
make_figures.py, make_figures_ml.py, build_report.py	figures and this report
verify.py	independent checks (Appendix B)
inspection_app/index.html	digital visual inspection application (offline, single file)
out/data/*.csv	baseline year, scenario library, NDT surveys, visual inspection, 18-year monitoring record

Requirements: Python 3 with numpy, scipy and matplotlib only. Run the five `run_*.py` scripts in order, then `verify.py`, `make_figures.py all` and `build_report.py`. The whole chain takes about an hour on a 16-thread laptop, most of it the 3600 vibration records of the library.

Appendix B. Verification checks

Table 27. Independent checks of the computed results

Check	Computed	Independent	Diff.	Tol.	
precast area (polygon vs pieces)	0.855 m2	0.855	0.0 %	0 %	pass
precast centroid height (trapezoid centroids approximated at mid-height)	0.912 m	0.9131	0.1 %	1 %	pass
sum of girder mid-span moments, 10 kN/m per girder	8149 kNm	8168	0.2 %	1 %	pass
mid-span deflection, inner girder	6.782 mm	6.792	0.1 %	3 %	pass
G1 LL moment: grillage vs Courbon (Courbon valid here: L/B = 2.06, stiff cross girders)	2457 kNm	2426	1.3 %	5 %	pass
Class A max mid-span moment (one train)	3172 kNm	3172	0.0 %	0 %	pass
C3 force after friction at mid-span	2220 kN	2220	0.0 %	0 %	pass
total elastic shortening loss at mid-span (hand: (n-1)/2n rule, mean age 20 d)	370.3 kN	355	4.3 %	25 %	pass
G1 soffit stress, quasi-permanent (0.9 Pm)	3.319 MPa	3.365	1.4 %	5 %	pass
G1 ULS moment capacity	1.647e+04 kNm	1.649e+04	0.1 %	1 %	pass
bearing movement per degC (healthy)	0.1924 mm/degC	0.198	2.8 %	5 %	pass
f1, twin vs simply supported beam formula (beam ignores end blocks, cross girders and bearing flexibility)	3.741 Hz	3.842	2.6 %	4 %	pass
level change for 10 % prestress loss (hand: 10 % of the elastic Pe camber x (1 + chi phi))	10.38 mm	10.25	1.3 %	15 %	pass
VW strain change for 10 % prestress loss	123.9 microstrain	123.9	0.0 %	2 %	pass
identified f1 from a simulated 30 min record	3.752 Hz	3.741	0.3 %	0 %	pass
identified f3 from a simulated 30 min record	8.837 Hz	8.815	0.3 %	0 %	pass
updated E_scale vs value used to create the data	1.055 -	1.06	0.5 %	6 %	pass
updated barrier vs value used to create the data	0.3598 -	0.35	2.8 %	6 %	pass
updated E_dyn vs value used to create the data	1.245 -	1.24	0.4 %	6 %	pass
novelty false alarm rate on healthy test data (99 % threshold from training data; small test sample)	0.03309 -	0.01	2.3 %	3 %	pass

Appendix C. Notation

Symbol	Meaning
P_j, P_0, P_e	jacking force, force after immediate losses, effective force after all losses
η	ratio of actual to expected effective prestress
$f_1 \ldots f_4$	first bending, first torsion, first transverse, second bending frequency
a_{RMS}, a_{max}	RMS and peak acceleration of a record
δ, ε	deflection, strain
MAC	modal assurance criterion
FDD	frequency domain decomposition
HCP	half-cell potential (mV vs copper-copper sulphate)
UPV	ultrasonic pulse velocity
VW	vibrating-wire (strain gauge)
CS	condition state 1 (good) to 5 (critical)
BCI	bridge condition index, 0-100
GPR	Gaussian-process regression
EWMA	exponentially weighted moving average